Recovering the Designer's Ledger: Inverse Hedonic Pricing of Card Costs in a Collectible Card Game

Jiahao Chen and Jiaxi Catherine Lee

***Abstract*—We present a method that recovers a card game designer's unpublished costing rules from printed cards alone, and a preregistered protocol that tests the recovered model on an unseen set. Treating cost as an inverse hedonic pricing problem, ZHIZHI (Zero-residual Hedonic Inversion, Zero-shot Hedonic Imputation) inverts effect prices from anchor cards differing in a single effect, so most prices are identified exactly, then imputes any card's value from its text without refitting; the value-to-cost ratio is the AMBER score. On *Riftbound* it fully prices 497 of 702 costed units and spells (median score 1.00, IQR 0.86–1.07) and exposes design regularities: a 50% rule for repeatable contingent value, free play-timing keywords, a same-target complementarity premium and near-free on-play effects on cheap units. In the blind test, the frozen model priced the next set's 60 cards with RMSE 1.73 C (registered threshold 1.84) and median score 1.00. Its misses yield nine costing conventions that, in a revision made after unblinding, lower held-out error from 1.40 to 1.27 C on the earlier sets and to 1.10 C on the new set; a language model guessing from card text (1.31 C) beats the frozen model and is level with the revised one.**

***Index Terms*—collectible card games, game balancing, inverse hedonic pricing, hedonic imputation, preregistration, reverse engineering, Riftbound**

I. Introduction

Every card in a collectible card game (CCG) carries a printed cost and a description of what it does. The relation between them is set by the designer from an internal costing model—how much a point of damage is worth, how much a drawn card, how much a keyword—that is never published. Informal analyses have fitted linear cost models to small card pools by least squares [1], [2], and the game-AI literature has learned to *predict* printed costs from card text with black-box models [3], [4]; none delivers the model itself: a price for each effect, identified from cards that isolate it, with the evidence behind it and a test on cards not yet seen.

This paper recovers such a model for *Riftbound*, the *League of Legends* trading card game published by Riot Games [5], from static card data alone. The designer's costing function maps an effect bundle $\mathbf{x}$ to a cost c; we observe many $(\mathbf{x}_i, c_i)$ pairs and wish to identify the latent price vector—the hedonic pricing problem of Lancaster [6] and Rosen [7] applied to a market with a single, deliberate price-setter. We solve it in two stages, which we call ZHIZHI (Zero-residual Hedonic Inversion, Zero-shot Hedonic Imputation).[1] Stage 1 inverts effect prices from anchor cards that differ in a single effect, so that most prices are identified exactly rather than fitted; Stage 2 imputes the value of every card from its parsed text with the frozen price table, without refitting—the operation that price-index statistics call *hedonic imputation* [8]. Our target is *unit fairness*—whether a single card's statistics and effects are commensurate with its cost—rather than rule fairness or value fairness (whole decks under hidden information), and our measure of it is the AMBER score (Attribute-Matched Budget-Efficiency Ratio): the ratio of a card's imputed value to its printed cost, with repeatable triggers booked at the half activation the designer implicitly charges for (Section III-D) and 1.00 as par. The object recovered is the designer's cost logic, not competitive strength: a card priced at par can still be strong or weak in play because of synergy, tempo or metagame. The paper follows one line: the recovered model passed a preregistered blind test on the next set; the misses of that test, read back into the earlier sets, yield conventions the frozen rules lacked; and the revised model, exploratory until registered in turn, is the most accurate version to date.

A. Contributions

1) **A method and a validation protocol**: ZHIZHI, simulation-free recovery of designer prices via zero-residual anchor pairs and zero-shot imputation for every card, with a confidence tier per price and closed-form mechanism models—order-statistics pricing of selection [9], [10] and asymmetric mental accounting of hand cards [11], [12]; and a preregistration protocol for testing any such model: freeze the table, register hypotheses and thresholds before the next set's preview season, price it blind, unblind by script, and read the misses back as candidate conventions for the next registration.
2) **The AMBER score and its validation**: an interpretable per-card balance metric with cross-predictions between separately calibrated parameters, a nested ablation under leave-one-out cross-validation and a parameter audit; over 497 fully priced units and spells the score has median 1.00 and interquartile range 0.86–1.07.
3) **Empirical design regularities**: a 50% rule for repeatable contingent value against a zero booking of most one-off conditions, free timing keywords, a same-target complementarity premium and integer-rounding absorption on cheap units.
4) **The test and what it taught**: under the protocol, the frozen model priced the fifth set blind with RMSE 1.73 C

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[1] ZHIZHI and AMBER are also the names of the authors' two cats.

against a registered threshold of 1.84 C and median nominal score 1.00 (Section VIII). Its misses, read back into the earlier sets, yield nine costing conventions that, in a revision made after unblinding, lower leave-one-out error from 1.40 to 1.27 C and the new set's error to 1.10 C (Section IX, exploratory); a language-model baseline beats the frozen model and is level with the revised one (Section VIII-C).

II. Related Work

Game balance and CCG analytics. Game balancing is a broad design goal [13], usually measured through play: restricted-play agents [14], evolutionary search over game parameters [15], and for CCGs evolutionary deck construction and playtesting [16], [17], evolving card attributes toward matchup parity [18] and Monte Carlo search for *Magic: The Gathering* (MTG) [19]; Hoover et al. [20] survey the field. These evaluate balance *ex post*; we recover the *ex ante* costing rules. The earliest explicit attempt we know of is Bursztein's least-squares fit over the first *Hearthstone* set [1], [2], without an error metric; Jin [21] proposes strength functions for the same game. Closest in scale, Zuin and colleagues [3], [4] train text-representation networks and gradient-boosted trees to recommend the mana cost of an MTG card, and Bertram et al. [22] learn card representations that predict draft picks; both learn a mapping from card to cost or strength, neither an explicit price per effect with provenance. Summerville and Mateas [23] generate MTG cards with sequence models, and Ham [24] treats rarity as a balance lever.

Hedonic pricing and mental accounting. Lancaster [6] reframed goods as bundles of characteristics and Rosen [7] formalised their implicit prices. We keep the decomposition but replace the market with a single designer: prices are set by rule, so anchor pairs identify them exactly, and costs are integers, so rounding is a first-order source of residual variance. Thaler's mental accounting [11], [12] supplies the language for hand cards, whose stock is priced asymmetrically while the act of playing is not; hedonic imputation [8] is the operation Stage 2 performs card by card.

Selection value and small-sample statistics. The value of keeping the best of several uncertain alternatives is classical [25]–[27]; for normal candidates the expected maximum of n draws [9], [10] prices "look at N, keep one" and modal effects with one calibrated dispersion. Because most parameters rest on 3–15 cards, we rely on medians and the Hodges–Lehmann estimator [28], minimax fits beside least squares [29], the Mann–Whitney test [30], permutation tests and hold-out prediction [31], following the programme of robust statistics [32].

III. Game Background and Problem Formulation

A. Riftbound in Brief

Riftbound [5] is a two-resource CCG. Each card has an *energy* cost and may additionally require *power*, shown as rune icons of one of six *domains*; runes are channelled from a separate deck and can be *exhausted* or *ready*. *Units* have *might*, fight at *battlefields* and score points by *conquering* and *holding* them; *spells* resolve once; *gear* is a reusable permanent; *legends* and battlefields have no cost. Timing keywords govern when a card may be played: Action cards on the owner's turn or during *showdowns* (combat), Reaction cards "any time, even before spells and abilities resolve", Hidden cards face-down and later as reactions. Other keywords used below: Assault/Shield ($+X$ might while attacking/defending), Ganking (may move between battlefields), Deathknell (effect when the unit dies), Flow (play from trash for a Flow cost), Empower, Vision/Predict (look at the top card(s)) and Level/XP.

TABLE I
Card Pool and Valuation Coverage (Rows Copied from the Full Table in App. S-III)

Class	n	Full	Partial	Deferred
Unit	469	359	51	59
Spell	233	138	19	76
Costed (η defined; ranked, ablated)	702	497 (70.8%)	70 (10.0%)	135 (19.2%)
Gear (expected-uses model; not ablated)	107	69	19	19
Legend (per-use net value; not ranked)	49	46	—	3
Battlefield (per-trigger yield; not ranked)	64	61	—	3

One buff-token reminder card is excluded. Deferral rules with counts, and the composition of the fully priced, partially priced and deferred units and spells by cost, type, set, domain and rarity, are in App. S-III.

B. Data

The card pool was exported from the Rift Atlas database [33]: 923 cards with cost, might, domain, set, keywords and rules text across four sets—*Origins* (OGN), *Spiritforged* (SFD), *Unleashed* (UNL), *Vendetta* (VEN)—and the starter set *Proving Grounds* (OGS), from the Simplified Chinese release, which preceded the English one; card names follow the official English printings and collector numbers are given so every anchor can be checked. Rules text was parsed into effect annotations and priced per effect type (Table I). The fifth set, *Radiance* (RAD), was used only for the preregistered blind test of Section VIII; nothing in the price table was estimated from it.

Gear is priced with one extra parameter, the number of uses the designer expects; because that parameter is not tested by the ablation of Section VII-B, the statistics in this paper cover the 497 units and spells only, and legends and battlefields, which have no cost, get per-use yields rather than rankings. Effects whose value depends on quantities outside the card are excluded from pricing (effects that scale with might, counters and control theft, effects on all or an indefinite number of targets); symmetric effects are booked at zero. Ban status is not a pricing criterion (Section VI-D); three banned spells and two banned battlefields set aside under an earlier version of the deferral rules stay outside the statistics so that the preregistered frozen model is unchanged. Deferral is mechanical, but not random: fully priced units and spells are cheaper than partially priced and deferred ones (median C_{tot} 4 against 5 C), more often common (43% against 11–13%) and less often spells (28% against 56%), so the median AMBER score of 1.00 describes the priced population, in which expensive rare spells with pool-wide, might-dependent or counter effects are under-represented (App. S-III).

C. Problem Statement

Let card i have printed energy E_i, power P_i, might m_i (zero for spells) and an effect vector $\mathbf{x}_i \in \mathbb{R}^K_{\geq 0}$ counting K effect types, and let $\mathbf{v} \in \mathbb{R}^K$ be the designer's latent prices. The balance condition, up to integer rounding and deliberate deviations, is

$$m_i + \mathbf{v}^\top \mathbf{x}_i \;\approx\; C_{\mathrm{tot}}(E_i, P_i), \tag{1}$$

where C_{tot} converts the two resources to the common unit. Given $\{(E_i, P_i, m_i, \mathbf{x}_i)\}$, identify C_{tot} and $\mathbf{v}$ and characterise the residuals. Two features distinguish this from a regression: identification is often *exact*, a pair of cards differing in a single attribute pinning its price without error, so anchor pairs are preferred to global fits; and the response is integer-valued, so rounding contributes a structured, one-sided residual of less than 1 C that must be modelled rather than averaged away.

D. The AMBER Score

The value $V = m + \mathbf{v}^\top \mathbf{x}$ is evaluated at the card's *base* scenario, conditions unmet and repeatable triggers fired zero times, giving V_0; V_1 is the value after exactly one trigger. The *nominal* ratio is $\eta = V_0 / C_{\mathrm{tot}}$, and the AMBER score books a repeatable trigger at a fixed fraction of an activation,

$$\eta^\star \;=\; \frac{V_0 + k\,(V_1 - V_0)}{C_{\mathrm{tot}}}, \tag{2}$$

with $k = 0.5$ for a repeatable trigger, $k = 0.25$ for a trigger that carries an independent condition, and $k = 0$ otherwise.[2] The factor 0.5 applies to units whose repeatable trigger has a positive single-trigger increment, 162 of the 497 fully priced cards, and is halved for the 17 of them whose trigger sentence requires, beyond the triggering event, an independent state or outcome; spells, gear and all other cards have $k = 0$ (evidence and circularity in Section VI-A). Conditional bonuses ("if …, then …") stay at zero in the base scenario and at full value in a *full-trigger* reference reported alongside but not used for ranking—a tested convention (Section VII-B). Throughout, $\eta = 1.00$ is par—a unit of measurement, not an average: anchor cards return 1.00 by construction, and $\eta > 1.00$ denotes a nominal budget surplus at the unit-fairness level, not that the card is strong in play.

Notation. C_{tot}: printed cost in C; V_0, V_1: base and one-trigger values; η, $\eta^\star$: nominal ratio and AMBER score; $\eta_{(t)}$: ratio with a repeatable trigger booked t times; $B(c)$: might budget (Section IV-C); k, λ: trigger and DEATHKNELL factors; n^*, f^*: implied activations and implied realisation fraction of a one-off bonus of value Δ_{cond} (Section VI-A).

IV. METHOD

Stage 1, zero-residual hedonic inversion (Sections IV-A–IV-F), recovers the price table: the unit of account, the linear additive core with its anchor pairs, the might budget, the mental-accounting and order-statistics mechanisms and the fit criteria.

[2]In v1 of the preprint the ratio was unnamed; in v2 and in hypotheses H1–H8 of the preregistration, "AMBER score" denotes the nominal score η. Since v3 the AMBER score uses $\eta^\star$.

Stage 2, zero-shot hedonic imputation (Sections IV-G–IV-H), evaluates every card from its parsed effect vector with the frozen table and no refitting; a card all of whose effects are priced is fully priced, one with an unpriced effect is partial, and a card caught by a deferral rule is deferred.

A. Unit of Account and Resource Conversion

All values are in cost units C, with 1 energy = 1 C. The exchange rate for power is identified by three independent card equations—*Hextech Ray* (OGN-009), *Falling Star* (OGN-029) and *Punch First* (SFD-097), each paying a mix of energy and power for effects already priced in C by energy-only cards—and all three solve exactly to 1 power = 2 C, corroborated by *Vengeance* (OGN-229) (4 energy, 2 power) and *Blast of Power* (OGS-012) (6 energy, 1 power), which have the same total under the rate and the same effect (kill a unit). Hence $C_{\mathrm{tot}} = E + 2P$.

B. Linear Additive Hedonic Core (Stage 1: Inversion)

The value of a card is the sum of its priced components,

$$V = m + \sum_{k=1}^{K} v_k x_k, \tag{3}$$

with modal choices, conditional effects and other non-additive components modelled separately. Prices v_k are identified by *anchor-pair elimination*: two cards that differ only in effect k give v_k directly; cards differing in two effects give a two-equation system, and so on. *Incinerate* (OGS-003) (2 C, deal 2) and *Final Spark* (OGS-022) (8 C, deal 8) give 1 C per point of damage with a zero intercept; *Consult the Past* (OGN-083) (4 C, draw 2) and *Progress Day* (OGN-114) (8 C, draw 4) give 2 C per card, again through the origin. A price carries the confidence tier *high* when two or more mutually independent cards agree, *medium* when it rests on a single method or few samples, *medium–low* when a single anchor supports it, and *low* when it rests on an assumption. Because anchor cards return $\eta = 1.00$ by construction, agreement among anchors is not itself evidence of balance.

C. Piecewise-Linear Might Budget

Vanilla units (no effect text) satisfy might = cost up to 5 C, after which might grows faster than cost; every vanilla unit satisfies might = $\lfloor B(c) \rfloor$ for

$$B(c) = \begin{cases} 1.1\,c, & c \leq 5, \\ 5.5 + 1.33\,(c-5), & c > 5, \end{cases} \tag{4}$$

in C, one might being worth 1 C. Under the floor convention the slopes are identified only up to intervals ($[1.00, 1.20)$ and $[1.25, 1.375)$), within which $\lfloor B(c) \rfloor$ is unchanged at 1–9 C; the places that use the unrounded slack $B(c) - c$ (Sections IV-H and VI-C) are marked as such (App. S-II). $B(c)$ is the total value the designer expects a c-cost unit to deliver; the η denominator remains C_{tot}.

D. Mental Accounting of Hand Cards

In the designer's ledger a played card does not "cost a card": a per-card hand charge would appear in anchor pairs as a common non-zero intercept, and both the damage pair and the draw pair pass through the origin. We read this as a mental-accounting model [11] that treats the per-turn draw as an exogenous endowment: the *act* of playing converts a hand slot into board value rent-free, and only effects that change the *stock* of hand cards are priced, asymmetrically, at the margin—draw $1 = +2.00$ C, discard (the lowest-quality card in hand) $= -1.00$ C; the 1.00 C gap is the reserve price of a hand card (App. S-II).

E. Estimation and Inference

For parameters supported by several cards we compute both the least-squares and the minimax solution and choose a value between them; ratio-type parameters use regression through the origin; medians are the primary location statistic [28]; cards that deviate markedly from their peers are excluded as anchors and used only as bounds. To test whether an attribute is priced, cards are grouped by it and the mean and 95% interval of (cost − priced effect value) are computed, and parameters fitted on one group are used to predict a disjoint group (Section VII). Throughout we treat 0.3 C as the *resolution* of the model: a working threshold, not an estimated quantity—about a third of the 1 C cost step, the size of the disagreements the data leave unresolved between estimators of the same parameter (least squares against minimax: PREDICT 0.385 against 0.40 C per card, λ 0.46 against 0.54, gear removal 2.75 against 2.875 C) and the typical half-width of the bootstrap intervals of multi-anchor parameters. Differences below it are reported but not read as evidence.

F. Order-Statistics Model of Selection Premia

PREDICT, look-N-take-1 and "choose one of X" effects let the player keep the best of several uncertain options. Modelling option i as $\mu_i + \sigma Z_i$ with standardised noise, the premium from choosing is $\Delta = \mathbb{E}[\max_i(\mu_i + \sigma Z_i)] - \max_i \mu_i$; for equal means and normal noise, $\Delta \approx 0.564\sigma$ ($n = 2$), 0.846σ ($n = 3$), 1.029σ ($n = 4$) [9], [10]. The dispersion is calibrated once per mechanism from cards whose other components are independently priced; the same model gives the expected value of the *worst* of n cards, which discard effects remove.

G. Discrete Scenario Binning (Stage 2: Imputation)

For repeatable triggers and count-dependent effects we impose no external probability or deck model: value and η are reported in bins of 0, 1, 2 and 3 triggers, with the base scenario at zero and the AMBER score at the half activation of Section III-D; conditional effects are reported at two bounds, base and full-trigger.

H. Separating Effect Value from Might

For spells, effect value is C_{tot} minus all other priced components. For units above 5 C it is $B(c) - m -$ (other priced components), on the assumption that the high-cost scale discount applies to the whole value budget rather than to might alone: 19 units carrying a single on-play effect, separated this way, yield effect values of $1.10\times$ (resource effects) and $1.12\times$ (removal) the price of the same effect on a spell, the excess being the fractional budget slack, whereas under the alternative they would be worth only $\approx 0.8\times$ spell price. For units at or below 5 C the sub-1 C slack is rounding and is not credited; effect value is simply $c - m$ (App. S-II).

TABLE II
PRINCIPAL PRICES RECOVERED BY ZHIZHI (STAGE 1), IN C. ROWS COPIED FROM THE FULL PRICE TABLES OF APP. S-I; H = TWO OR MORE INDEPENDENT CARDS AGREE

Effect	Price	Conf.	Identifying evidence
1 power (rune icon)	2.00	H	Hextech Ray, Falling Star, Punch First (exact)
1 might	1.00	H	5 vanilla units; 18 further units with only zero-priced keywords
1 damage	1.00	H	Incinerate vs. Final Spark; no premium for battlefield-only targets
Draw 1 / discard 1 (own hand)	2.00 / −1.00	H	Progress Day, Consult the Past; Lunar Boon, Chemtech Enforcer
Kill a unit (unrestricted)	8.00	H	Vengeance, Blast of Power; also the overflow cap for damage and $-X$ might
Return enemy / own unit to hand	6.00 / −1.00	H	Rebuke; Retreat, Star-Crossed
Stun an enemy unit (any / restricted)	4.00 / 2.00	M–L/H	Rune Prison (single anchor); Facebreaker, Thwonk!, Heroic Charge
Score 1 point	11.50	H	6 cards, mean = median
SHIELD X, ASSAULT X	0.50 per point	H	Half price (conditional on attacking/defending)
ACTION, REACTION, REPEAT, HIDDEN, ACCELERATE, DEFLECT, AMBUSH, TANK	0	H	Multiple cards each; timing analysis in Section VI-B

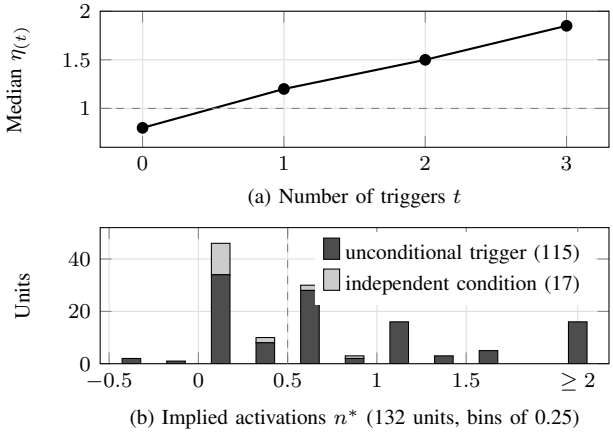
V. RESULTS: THE RECOVERED PRICE TABLE

A. Base Prices

Table II lists the principal prices with the anchors that identify them; the full tables (33 base prices by confidence tier, selection effects, target-restriction discounts, extended mechanisms, low-confidence prices) are in App. S-I. The price of a point of damage does not depend on whether the target must be at a battlefield; ASSAULT X and SHIELD X are worth 0.50 C per point because they apply only when attacking or defending; unconditional kill (8.00 C) serves as the overflow cap for single-target damage and "$-X$ might this turn", since only 9 units (1.9%) have might ≥ 9; and scoring one point is 11.50 C from six cards whose mean equals their median, a design-side threshold rather than an in-game exchange rate (Section X). Restricted versions of an effect are priced as a fraction of the unrestricted price (0.50 for a state or position; 0.55 for kill with might ≤ 3), restricted kill sitting *above* equivalent damage: *Soul Harvest* (UNL-159) is printed at 4 C where 3 damage costs 3 C, an instance of nominal η exceeding effective η.

B. Three Representative Mechanism Identifications

Selection (order statistics). PREDICT N does not change hand size; it improves the quality of subsequent draws. Jointly fitting *Clairvoyance* (VEN-056) (7 C, Predict 5 then draw 2), *Apprentice Mage* (VEN-047) and the discard price gives 0.385 C per card (least squares) and 0.40 C (minimax); we adopt 0.40 C, which also prices look-2/3/5-take-1 at 2.55/2.83/3.12 C and a two-mode "choose one" premium of $+0.8$ C. *Deathknell.* Booked in full, Deathknell units would systematically exceed their cost; solving $\lambda = (\lfloor B(c) \rfloor - m)/V_{\mathrm{DK}}$ on the five single-Deathknell units gives 0.46 (least squares), 0.54 (minimax) and 0.50 (median and mean), and we adopt $\lambda = 0.50$. *On-play effects (cost-controlled regression).* Effects realised through same-turn combat ($+X$ might, stun) are worth less on a unit, which enters exhausted outside showdowns, than on a spell, and cheap units price on-play effects at a small fraction regardless of type. The two are confounded (seven of nine initial combat-trick anchors cost ≤ 4 C), so over 25 units with a single on-play effect priced independently elsewhere we regress the recovered fraction of spell price on two indicators: low cost (2–3 C) gives -0.63 ($p < 0.001$) and combat-trick type -0.27 ($p = 0.057$), with a best multiplicative fit of $0.25\times$ and $0.55\times$; optional-additional-cost anchors give 0.50 for three units, so we adopt 0.50 for combat tricks and 0.30 for cheap units. The remaining identifications follow the same pattern (App. S-II).



(a) Number of triggers t

(b) Implied activations n^* (132 units, bins of 0.25)

Fig. 1. The 50% rule and its limit. (a) Median binned value/cost ratio $\eta_{(t)}$ of the 162 units with a repeatable trigger, by number of triggers booked (dashed line: par, 1.00); the curve crosses 1.00 near half a trigger, and the AMBER score (Section III-D) adopts this 0.5-activation baseline as its ranking convention. (b) Implied activations $n^* = (C_{\mathrm{tot}} - V_0)/(V_1 - V_0)$ for the 132 units whose single trigger is worth at least 1 C (bins of 0.25, left-closed; the dashed line marks 0.5). The distribution is spiky: 40 units sit exactly at 0 (9 of the 17 conditional triggers among them), 20 at 0.5 and 15 at 1. Panel (c), the implied realisation fraction f^* of the 46 one-off conditional bonuses, is in App. S-III.

C. Same-Target Complementarity Premium

One class of spells departs consistently from additivity. Among the eight fully priced spells in which "move a friendly unit" is the only unknown, four whose other effect targets a different unit, or that have no other effect, all imply exactly 1.00 C (*Flash* (OGS-011), *Shuriken Flip* (VEN-140), *Void Assault* (UNL-202), *Tactical Retreat* (UNL-175)); three that reposition a friendly unit and then boost the *same* unit all imply exactly 2.00 C (*Ride the Wind* (OGN-173), *Showstopper* (OGN-270), *Resonating Strike* (VEN-034)). With *Vault Breaker* (UNL-010) and *Perfect Execution* (VEN-012), five cards carry 1.0–1.3 C over their additive value and none of this shape is priced additively, whereas four spells that split positioning and enhancement across different units show zero residual. We therefore book a *same-target complementarity premium* of $+1.00$ C for a spell that both positions a friendly unit and enhances it for combat—an interaction term tracking target identity rather than single resolution (an "atomicity" reading would predict the same premium for the split-target spells), applied to spell base values only and, with five supporting cards, treated as a hypothesis (App. S-II).

VI. Results: Latent Design Regularities

A. The 50% Rule for Contingent Value

Across the 162 units with a repeatable trigger, the median binned ratio $\eta_{(t)}$ at $t = 0$, 1, 2 and 3 triggers is 0.80, 1.20, 1.50 and 1.85, crossing 1.00 at about half a trigger (Fig. 1(a)): the designer prices a repeatable benefit as if it fired half a time, and the same factor recurs in independent fits (DEATHKNELL at $\lambda = 0.50$, combat-trick on-play effects at 0.5). *Implied activations.* The number of triggers the printed cost pays for, $n^* = (C_{\mathrm{tot}} - V_0)/(V_1 - V_0)$, has median 0.500 (bootstrap 95% interval 0.354–0.500) over the 132 units whose single trigger is worth at least 1 C, with 78.8% between 0 and 1 and spikes at 0 (40 units), 0.5 (20) and 1 (15) (Fig. 1(b)). *Conditional triggers.* The 17 whose trigger sentence requires an independent condition (Section III-D) have median n^* 0.00 (95% interval 0.00–0.26) against 0.50 (0.50–0.67) for the other 115 (Mann–Whitney $p = 0.003$, descriptive): the designer charges almost nothing for a repeatable effect gated by a further condition, and the AMBER score books these at a quarter activation. *Conditional bonuses.* The rule does not extend to one-off "if …, then …" bonuses. Among the 46 fully priced cards with such a bonus worth at least 1 C (selected by rules text before costs were compared), the fraction of the conditional value that the printed cost pays for, $f^* = (C_{\mathrm{tot}} - V_B)/\Delta_{\mathrm{cond}}$, has median 0.08 (95% interval 0.00–0.50) and a two-humped distribution (App. S-III): 20 cards exactly at 0, 16 between 0.5 and 0.8. Where the condition is one the player arranges, the median is 0.67 ($n = 9$, against 0.00 for the other 37; $p = 0.052$); we record as a hypothesis that the designer books contingencies the player controls at about one half and contingencies on the game state at zero, and the base scenario keeps conditional bonuses at zero (Section VII-B). *Circularity.* The 0.5 factor was estimated on these 162 units, so their median AMBER score of 1.00 (0.80 nominal) is a consequence of the construction, not evidence of balance, and the 17 conditional cards were tagged, by rules text, after the ranked list had been reviewed. The out-of-sample tests of the factors and of the zero booking were registered as H9 and H10 (Section VIII).

B. Play Timing Is Free; Choice Is Priced

For 47 spells whose effects are fully priced, grouping by timing keyword and comparing cost minus priced effect value shows no charge: the most flexible group (REACTION, $n = 12$) has the *lowest* mean (-0.37 C), the 95% interval for all keyword-bearing spells is -0.61 to $+0.36$ C, and same-effect contrasts show no difference (App. S-III). A wider play window is a strict advantage and in other CCGs usually carries a discount; here timing is a free qualitative tag, plausibly kept free to encourage interactive, second-mover play. By contrast, modal cards pay ≈0.8 C above their best mode, and selection premia are concave in the number of options. The one exception is a this-turn might change or stun on a spell *without* any timing keyword, booked at 0.5 on the single calibrating card *Onslaught* (VEN-081) (low confidence).

C. Cheap Units Get On-Play Effects Almost Free

Stratifying the 25 single-on-play-effect units by cost shows a sharp layer: 2–3 C units price the effect at roughly 30% of spell price regardless of type (best fit 0.26 with the combat-trick factor fixed at 0.5), while units at 4 C and above pay full price (13 cards, median 1.00). Integer might is the mechanism—*integer-rounding absorption*: attaching a 1–4 C effect to a cheap unit removes 0 or 1 might, so most of the effect is a gift, whereas the same effect sold as an optional additional cost on a cheap unit is priced at 0.5 or more. Keywords that a 2–3 C unit carries natively are rounded the same way when their total price is below 1 C (seven of nine such units have might equal to cost; keywords worth 1 C or more are paid for in might by five of six), so the model books them at 0.30 of their price; the 3 C boundary follows from the budget slack under the slope 1.1 (App. S-III).

D. Further Regularities

XP is priced at both ends but hidden by rounding. Over the 18 XP-gaining units the gain-side price is 0.38 C per point (adopted 0.40, close to the spend-side 0.50); 12 cheap units imply ≈ 0 because 0.4 C of XP is absorbed by rounding. *Flow is booked on the first play.* The printed-side premium for the keyword is small (0.2 C adopted, range 0–0.6 C; six of seven REPEAT spells carry a premium of exactly 0) and vanishes once the Flow cost exceeds the printed cost by 1. *Scenario effects at about six tenths.* Unconditional gear removal is priced from four anchors with median 2.75 C, and one free EMPOWER at ≈ 2.4 C (median Empower cost 4 C $\times 0.6$; *Wild Claw* (VEN-089) recovers 2.36 independently): effects the opponent does not always offer a target for, booked below full value. *Deathknell booking by set: a hypothesis.* Per-set fits of λ trend upward with release order (0.53, 0.44, 0.69, 0.86), but the slope is $+0.093$ per set (s.e. 0.060; permutation $p \approx 0.10$) and no test rejects a common λ, so the model uses 0.50 for all sets. *Ban status is not evidence of single-card excess.* Twelve cards are banned in Standard as of 24 September 2026; the seven costed ones span the AMBER scale rather than its upper tail (*Stacked Deck* (OGN-183) scores 2.83, the same as the unbanned *Lightning Rush* (VEN-156); *Stealthy Pursuer* (OGN-177) 0.75), and what the bans share—loops, deck-thinning consistency, tempo, first-player effects—lies in the value-fairness layer the score does not measure (App. S-III).

TABLE III
CROSS-PREDICTIONS BETWEEN SEPARATELY CALIBRATED PARAMETERS (ROWS MARKED C: CONSISTENCY CHECKS, NOT HOLD-OUTS; FULL VERSION WITH CALIBRATION SOURCES IN APP. S-III)

Quantity predicted (calibrated from)	Prediction	Observed
(C) Discard the lowest-quality card in a hand (order-statistics model, jointly fitted dispersion)	0.83–1.04 C	1.00 C (Lunar Boon, Chemtech Enforcer; Bewitching Spirit for the opponent's hand)
Zero hand-rent intercept (damage pair Incinerate / Final Spark: 1 C per point, intercept 0)	Other effect lines pass through the origin	Draw: Consult the Past (4 C, draw 2), Progress Day (8 C, draw 4) give 2 C per card, intercept 0
(C) 1 power (three spell equations)	2 C	Channel 1 exhausted rune = 2 C (Mobilize, Catalyst of Aeons)
One free Empower (median Empower cost $\times 0.6$)	2.4 C	Wild Claw recovers 2.36 C
Ready 2 C, buff 1 C, return-from-trash 2 C (individual anchors: Upstage Comedy, Wallop; Morbid Return)	base $\eta = 1.00$	Call to Glory (OGN-207), Confront (OGN-129), Rally the Troops (SFD-166), Here to Help (SFD-111), Undying Loyalty (UNL-168) all 1.00
Opponent chooses the worse mode (modal model, expected minimum)	3.84 C	Shakedown printed 4 C
Remove damage + recall + exhaust bundle (Zhonya's Hourglass, Tactical Retreat)	≈2 C	Soraka, Wanderer: cost − might = 2 C for the same bundle as a trigger
May be played directly to a battlefield (Sneaky Deckhand, Rengar, Pouncing)	+1 C	Ocean Drake 1.00 C, Sai Scout 0.60; Dragon Roost battlefield yield 1 C

VII. VALIDATION

A. Cross-Predictions and the Distribution of the Score

The model's credibility rests mainly on predicting cards that were not used to calibrate the relevant parameter (Table III); each row is read as prediction against observation at the ≈0.3 C resolution, and rows marked C are consistency checks between separately anchored prices rather than hold-outs. Every hold-out row lands within the resolution: the zero intercept of the damage pair predicts that draw cards are priced through the origin, the free-Empower value matches Wild Claw, and the opponent-chooses variant of the modal model predicts Shakedown to within 0.16 C.

Over the 497 fully priced units and spells, the AMBER score $\eta^\star$ has mean 0.972, median 1.00 and interquartile range 0.86–1.07; 62 cards exceed 1.17, 11 reach 1.50 and 36.2% sit exactly at 1.00. The nominal ratio has mean 0.911, the same median and interquartile range 0.75–1.00. Anchor cards score 1.00 by construction; excluding the 146 of them leaves 351 cards with median 1.00 and interquartile range 0.80–1.09, and for the 162 trigger units the return to a median of 1.00 is partly a construction of the 0.5 factor (Section VI-A). The table thus reproduces the designer's pricing for the central mass of the pool, with a left tail of conservatively priced cards and a short right tail (histogram and the fifteen highest scores in App. S-III).

B. Ablation and Trigger Booking under Leave-One-Out

Table IV(a) compares four nested models—A, the linear additive core; B, A + mental accounting of hand cards; C, B + order-statistics selection premia; D, C + cost-controlled on-play adjustments and keyword rounding (the full model)—in-sample and under leave-one-out cross-validation [31], in which each card is held out, every anchor-derived price is re-estimated from the remaining cards with the same statistic and chaining, and the card is then predicted; sixteen cards whose price is identified by that card alone are dropped, leaving 481. Only layer D reduces error materially: in-sample RMSE 1.537 to 1.504, with 22 of the 26 units it changes moving closer to their cost, and the gain survives cross-validation almost intact (1.574 to 1.541), so the low-cost and combat-trick factors are not artefacts of their calibration cards. Because 436 of the 497 cards take the same value under all four models, panel (b) repeats the comparison on the cards each layer changes: D lowers RMSE on its own cards from 1.38 to 0.45 C out of sample with an interval that excludes zero, whereas B and C stay indistinguishable from the layer below even where they act ($p \geq 0.61$); whether the designer uses the hand-card and selection rules cannot be decided at this sample size and 1 C granularity, and we retain both layers for interpretability and because their cross-checks hold. The overall level of error comes from what no layer addresses: base values exclude triggered and conditional effects, so residuals are negative on average (-0.57 to -0.63 C) and R^2 stays between 0.66 and 0.68.

Panel (c) tests the trigger factors of Section III-D on model D under leave-one-out, each version adding $k(V_1 - V_0)$ to the held-out prediction and each pair compared on the squared errors of the cards whose k differs (paired bootstrap, sign-flip permutation and Wilcoxon tests). Both steps of the AMBER convention improve prediction card by card: uniform booking at 0.5 brings 95 of the 162 units closer to their cost and lowers the leave-one-out RMSE over all 481 cards from 1.541 to 1.438 C, and the conditional halving brings 14 of its 17 cards closer and lowers it to 1.400 C, with intervals excluding zero on the changed cards and on all 481. Two qualifications: leave-one-out re-estimates the effect prices in V but not the factors themselves, so the table asks whether the fixed factors improve prediction card by card and is not an out-of-sample test of them (that test is H9 of Section VIII); and the conditional gain is concentrated in four high-increment cards that account for 90% of its reduction in squared error. The same test on one-off conditional bonuses finds no factor to adopt (booking Δ_{cond} at 0.5 brings 23 of 53 cards closer and 30 further; booking it in full is significantly worse), so the AMBER score keeps $k_{\text{cond}} = 0$.

C. Parameter Review

After the ablation, every price was re-examined against the cards that use it: for each of the 37 multi-anchor or layer parameters we tested whether the clean cards among them (base value equal to full-trigger value, 199 cards) show a one-sided residual in-sample and under leave-one-out, recorded how far the parameter moves when any one anchor is held out, and, for flagged parameters, compared the leave-one-out error of alternative estimation rules with exact bootstrap intervals; a change was to be adopted only if it followed from the parameter's own anchors under the same statistic, exceeded the 0.3 C resolution and did not predict held-out anchors worse. No parameter met these conditions (App. S-III); the single-anchor prices for which a second card disagrees without a way to decide are listed among the low-confidence prices in App. S-I, and two methodological corollaries (never tune a parameter on the pooled residual of the cards that use it; read unit residuals against $\lfloor B(c) \rfloor$ for might and $B(c)$ for effects) in App. S-III.

VIII. Preregistered Blind Test on the Fifth Set

The model (rules v6 and the value table, with the scoring scripts) was frozen and registered on 24 September 2026, the day before the official preview season began; H9 and H10 were added in an addendum on 25 September, before any *Radiance* card was priced (Data Availability). This section reports the registered hypotheses exactly as the frozen scripts score them; Section IX reports what the set suggests beyond them.

A. Protocol and Deviations

Once the set was complete (10 October), the card list was split into a blind table (rules text, type, might, domain, keywords; 167 cards) and a cost table that was not opened. The blind table was priced card by card under the frozen rules, each card marked fully priced, partial or deferred, and the tags used by H3–H10 were assigned before any cost was seen. The pricing was done by a separate AI session (a Claude sub-agent; Acknowledgment) given the blind table, the frozen rules and the price book and instructed to open nothing else; its isolation rests on that instruction (Section X-A). Before unblinding, the first author reviewed the blind pricing and corrected four cards (one misreading, one card wrongly left partial, two omitted "can't move this turn" clauses booked at -1 C by analogy with exhausting a unit); the original file, the revised file and the list of changes were published together as version 3 of the preregistration record, with SHA-256 hashes checked before the cost table was read, and, as announced there, the revised file is the primary analysis with the original reported alongside. Unblinding was run by a script that read the cost table only after six integrity gates (DOIs, hashes of every input, the frozen scripts' stored self-check) had passed and called the frozen scoring scripts unchanged. Two deviations bear on the results (full list in the unblinding report): cards carrying a new keyword (Disarm, Show Off, Bomb, Deploy) were left partial, since the frozen model has no price for them; and *Mournful Spirit* (RAD-126), tagged as a single-Deathknell card during blind pricing, has a condition-laden Deathknell, so with it removed H7 has one card. Of the 122 units and spells, 66 are fully priced, 32 partial and 24 deferred; 60 of the 66 carry no low-confidence price and form the primary sample (the fully priced share, 54%, is lower than the old pool's 71% chiefly because Disarm alone leaves nine units partial).

B. Results

Table V lists every registered hypothesis. Both primary hypotheses hold under both files: the frozen model prices the

TABLE IV
VALIDATION UNDER LEAVE-ONE-OUT: NESTED ABLATION, ABLATION ON THE CARDS EACH LAYER CHANGES, AND TRIGGER / CONDITIONAL-BONUS BOOKING AS A PAIRED TEST (ROWS COPIED FROM THE FULL TABLES IN APP. S-III)

(a) Nested models, all fully priced cards (in-sample $n = 497$; leave-one-out $n = 481$)

	In-sample			Leave-one-out		
Model	RMSE (C)	MAE (C)	R^2	RMSE (C)	MAE (C)	R^2
A: linear core	1.538	0.910	0.664	1.575	0.957	0.650
B: A + hand-card mental accounting	1.538	0.910	0.665	1.575	0.957	0.650
C: B + order-statistics premia	1.537	0.910	0.665	1.574	0.957	0.650
D: C + on-play cost adjustments (full)	1.504	0.871	0.679	1.541	0.916	0.665

(b) Each layer on the cards it changes, leave-one-out

Layer	n	RMSE (C)	ΔRMSE [90% CI]	$+/-/=$	p
A→B	16	0.598→0.594	−0.003 [−0.03, +0.03]	8/7/1	0.95
B→C	20	1.083→1.048	−0.035 [−0.16, +0.10]	12/8/0	0.61
C→D	29	1.380→0.453	−0.927 [−1.30, −0.53]	25/3/1	<0.001

(c) Trigger and conditional-bonus booking under model D, leave-one-out (paired over the changed cards)

Comparison	n	$+/-/=$	RMSE (C), changed	ΔRMSE [95%], changed	ΔRMSE [95%], all 481	p_{perm} / p_{W}
Uniform 0.5 vs. nominal	162	95/66/1	1.793→1.518	−0.275 [−0.435, −0.102]	−0.103 [−0.171, −0.036]	0.002 / <0.001
Baseline vs. uniform	17	14/3/0	1.926→0.818	−1.108 [−1.751, −0.368]	−0.038 [−0.080, −0.008]	0.023 / 0.013
Conditional bonus at 0.5 vs. baseline (0)	53	23/30/0	1.316→1.209	−0.107 [−0.513, +0.336]	−0.011 [−0.055, +0.037]	0.67 / 0.61
Conditional bonus at 1.0 vs. 0.5	53	8/43/2	1.209→2.053	+0.845 [+0.534, +1.128]	+0.105 [+0.049, +0.180]	<0.001 / <0.001

(a) Residual $r_i = V_i - C_{\text{tot},i}$ on base value; A: linear core; B: + hand-card mental accounting; C: + order-statistics premia; D: + on-play cost adjustments (full model, with keyword rounding). Sixteen cards whose price is identified only by the card itself are dropped from the leave-one-out. (b) Cards whose value differs between the two layers; $+/-/=$: cards whose $|r_i|$ decreases, increases or is unchanged; CI: percentile bootstrap over cards (10,000); p: exact Wilcoxon signed-rank. (c) Squared error of $V_{\text{LOO}} + k(V_1 - V_0)$ against C_{tot}; *uniform*: $k = 0.5$ for all 162 trigger units; *baseline*: with $k = 0.25$ for the 17 conditional triggers (the AMBER score); last two rows add $k_{\text{cond}}\Delta_{\text{cond}}$ on the 53 predictable cards with a one-off conditional bonus. Leave-one-out RMSE over all 481 cards: nominal 1.541, uniform 1.438, baseline 1.400 C. In-sample rows, the per-trigger-type versions, the k_{cond} grid and the exact-hit (McNemar) counts are in App. S-III.

TABLE V
PREREGISTERED HYPOTHESES ON *Radiance*, SCORED BY THE FROZEN SCRIPTS (REVISED BLIND PRICING IS PRIMARY)

Hypothesis	Criterion (minimum n)	Revised pricing (primary)	Original pricing
H1 prediction error	RMSE of $r = V - C_{\text{tot}} \leq 1.841$ C (20)	$n = 60$: 1.734 C (90% bootstrap 1.372–2.065) — **supported**	$n = 56$: 1.773 — supported
excl. cards revealed before 25 Sep.	same	$n = 57$: 1.666 — supported	$n = 53$: 1.704 — supported
all fully priced (secondary)	same	$n = 66$: 1.707 — supported	$n = 62$: 1.742 — supported
post hoc: five extreme spells re-read under the revised rules (Section IX)	same	$n = 58$: 1.683 (1.307–2.026) — supported	—
H2 median nominal score	$0.90 \leq$ median $\eta \leq 1.10$ (20)	1.000 — **supported** (all samples, the post-hoc tier included)	0.938 — supported; excl. early reveals 0.875 — not supported
H3 vanilla units	might $= \lfloor B(c) \rfloor$ (1)	no vanilla unit — untestable	same
H4a / H4b combination spells	median r within ± 0.5 (2 each)	$n = 0$ / 0 — untestable	same
H5 draw rider	median r in $+0.5$ to $+1.5$ (3)	$n = 0$ — untestable	same
H6 single-anchor conflicts	record only	unrestricted stun: two cards (both low-confidence) imply 2.00 and 2.00 (current 4, nearest prestated value 3; the registered change condition is met numerically); others no sample	same
H7 Deathknell λ	median ≥ 0.65 trend; 0.35–0.65 common 0.5 (2)	$n = 1$ (Hunting Dog) — untestable	same
H8 timing keywords free	$\lvert\Delta\rvert < 0.5$ and 95% CI $\ni 0$ (5 per group)	11 vs 5 spells: $\Delta = +0.14$ (-1.50 to $+1.55$) — **supported**	10 vs 4 — untestable
H9a repeatable triggers	median n^* in 0.35–0.65 (10)	$n = 16$: 0.583 (90% 0.000–0.774) — **supported**	same
H9b conditional triggers	median $n^* \leq 0.35$ (5)	$n = 3$ — untestable	same
H9c 0.5 vs nominal	lower RMSE, paired 90% CI excludes 0 (10)	$n = 19$: 1.996 → 1.302 C, ΔRMSE -0.693 (-1.211 to -0.155) — **supported**	same
H10a / H10b one-off conditions	median $f^* \leq 0.25$; 0.5 not better (10)	$n = 5$ — untestable	same

H1, H2, H4, H5 and H8 use the primary sample (fully priced units and spells without a low-confidence price); H3, H6, H7, H9 and H10 use their tagged cards; H9 and H10 are identical under both files. "Excl. cards revealed before 25 Sep." is the sensitivity analysis named in the registration; under it H9a gives $n = 14$, median 0.583, and H9c $n = 17$, 1.790 → 1.041 C (-1.318 to -0.110), both supported. The post-hoc tier was added after unblinding (App. S-IV and S-V). For H7 the frozen script also counts Mournful Spirit ($n = 2$); it is not a single-Deathknell card (Section VIII-A). The remaining tiers are in App. S-IV.

new set with RMSE 1.734 C (MAE 1.15 C) against a threshold of 1.841 C, and the median nominal score is exactly 1.00. H8 and the two testable parts of H9 are supported; H3, H4, H5, H7, H9b and H10 have too few cards in this set to be tested. H6 records only: *Ki Burst* (RAD-028) and *One-Two Punch* (RAD-121), on which an unrestricted stun is the one unknown, both imply 2 C, half the frozen price (Section IX), which stays unchanged.

Three qualifications apply. H1 passes on the point estimate, as registered, but the 90% bootstrap interval reaches 2.07 C, so 60 cards cannot rule out a true error somewhat above the threshold (the out-of-sample error is 0.19 C above the leave-one-out benchmark of 1.541 C). Base values sit below cost on average (mean residual -0.71 C), as in the old pool; booking

repeatable triggers at half an activation closes nearly all of the gap: over the 60 cards the mean AMBER score is 0.989 (nominal 0.923), within 0.011 of par, and both medians are 1.00. And H9a is fragile: seven of its sixteen units have $n^* = 0$ and three exceed 1, and a tier added after unblinding on the first author's reading of RAD-104 gives 0.667 ($n = 15$), just outside the band; H9c, the paired comparison, holds in every tier.

Sensitivity. Three tiers are reported in Table V: the registered primary analysis, the registered exclusion of cards revealed before 25 September, and one post-hoc tier, added after unblinding, that re-reads the five low-cost spells responsible for most of the set's spread under the revised rules of Section IX (one card out of scope as champion-bound, one as count-dependent without a cap, two re-booked, one unchanged; $n = 58$): H1 1.683 C (1.307–2.026) and H2 1.000, no verdict changing. All tiers are in App. S-IV.

C. Baselines

The baselines in Table VI were added at revision and are not part of the registration. B0 and B1 are clearly worse than the preregistered model; B2, a ridge regression on machine-readable card-face features, cannot be separated from it (ΔRMSE −0.039 C, 90% interval −0.428 to +0.311). B3, a language model given the blind table and nothing else, is more accurate than the preregistered model: 1.307 against 1.734 C, paired 90% interval +0.236 to +0.601, and the same holds in each of five independent runs (RMSE 1.226–1.324 C; every interval excludes zero). The exploratory conventions of Section IX (last row of Table VI) bring ZHIZHI to 1.100 C (1.079 C on the 59 cards without the signature spell RAD-154), at least level with B3: against its first run the paired interval just excludes zero on the 59 cards (−0.238, −0.460 to −0.007) and includes it on all 60 (−0.207, −0.431 to +0.018), and against the mean of the five runs it includes zero on both. What separates the two is not accuracy but accounting: every ZHIZHI price is a sum of effect prices that can be read off and traced to anchor cards, and its rules can be registered before a set is revealed and tested after it.

D. The Fifth Set Among the Five

Fig. 2 places *Radiance* beside the four calibration sets: its median AMBER score is 1.00, like every earlier set, and neither its mean (+0.045, permutation $p = 0.36$) nor its median differs from the pooled earlier sets; what changes out of sample is the spread (14% of its fully priced cards exactly at 1.00 against 34–40%; interquartile range 0.77–1.17).

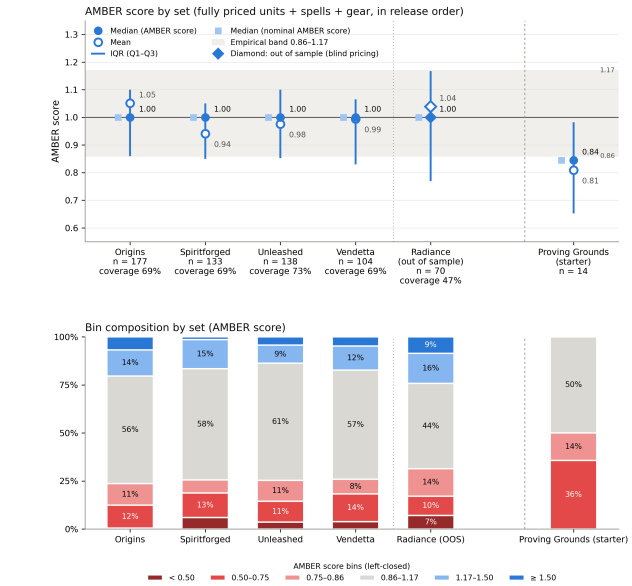


Fig. 2. AMBER score by set (fully priced units, spells and gear, in release order). Sets 1–4 are in sample; *Radiance* (diamond) is the preregistered blind pricing, out of sample, under the frozen rules. Top: median, mean and interquartile range against the empirical 0.86–1.17 band; bottom: composition by score band. The wider *Radiance* spread comes from five low-cost spells, none misread (Section X-A); under the revised rules of Section IX, applied post hoc, the Brown–Forsythe dispersion ratio (the set's mean absolute deviation from its median score over that of the leave-one-out earlier sets, 95% stratified bootstrap) falls from 1.59 [1.10, 2.18] to 1.28 [0.89, 1.78] ($n = 58$ against 465; App. S-IV and S-V).

IX. What the Fifth Set Adds (Exploratory)

The registered test having passed, this section turns its misses to account. Everything here was estimated after the costs were read, mostly on the set's own cards; none of it enters H1–H10, every item is marked † as a candidate for the next version of the model rather than a finding, and the full analyses are in App. S-V and S-VII.

New keywords.† Disarm N ("when I attack, give an enemy unit here $-N$ might this turn") is priced at 0.25 C per point from the four cards where it is the one unknown (joint least squares 0.26; leave-one-out refits 0.08–0.39). Show Off adds nothing by itself; the bonus it unlocks behaves like the one-off conditional bonuses of Section VI-A, booked at zero on four of five cards. A Bomb token is worth 1 C (low confidence), and Deploy costs nothing. "Can't move this turn" fits better at −2 C than at the −1 C of an exhausted unit (*Entrenched Boar* (RAD-026) is then priced exactly; the primary H1 error becomes 1.719 C).

Stun has fallen in price.† The frozen model priced an unrestricted stun at 4 C from *Rune Prison* (OGN-050) alone (medium–low confidence). Read by set, in the first set an unrestricted stun cost 4 C and a restricted one 2 C, and from the second set on every spell on which stun is the one unknown is priced at 2 C (*Thwonk!* (SFD-040), *Heroic Charge* (UNL-155), Ki Burst, One-Two Punch). Signature cards are not used as anchors: *Zenith Blade* (OGN-262), Leona's signature spell, sits 2 C below its cost at the frozen price and 4 C below at 2 C, and the 20 fully priced signature spells range from −3 to +2 C, each priced for its champion. On the nine remaining cards whose value changes, the staged price lowers RMSE from 1.16 to 0.90 C (paired 90% interval −0.77 to +0.29); on the four units that get worse the stun price and the activation count cannot be separated.

Reverse-engineered conventions.† The largest misses of the blind test fall on cards whose value depends on how many things are present or on an optional payment (*Party Crasher* (RAD-122), *Mordekaiser, Dominating* (RAD-137)),

TABLE VI
BASELINES ON THE FIFTH SET (ADDED AT REVISION, NOT PREREGISTERED): RMSE OF PREDICTED AGAINST PRINTED COST ON THE 60 PRIMARY CARDS, PAIRED DIFFERENCE FROM THE PREREGISTERED MODEL, AND LEAVE-ONE-OUT ERROR ON THE EARLIER SETS

Method	Definition	RMSE, $n = 60$ (90%)	ΔRMSE [90%]	p_{W}	LOO, $n = 481$
B0 constant	Mean cost of units, and of spells, over the earlier sets	2.805 (2.399–3.211)	-1.071 [-1.450, -0.709]	6.6×10^{-6}	2.629
B1 type × might	Unit cost = might; spell cost = spell mean	2.428 (2.025–2.806)	-0.694 [-1.043, -0.388]	1.4×10^{-4}	2.032
B2 ridge regression	49 machine-readable card-face features (type, domain, rarity, might, 26 keyword indicators, counts over the rules text); penalty chosen by ten-fold cross-validation	1.773 (1.486–2.042)	-0.039 [-0.428, $+0.311$]	0.369	1.568
B3 language model	Claude Opus 5.5 in a fresh session, guessing each cost from the blind table alone: no rules, no price book, no cost of any earlier card; the run fixed in advance as primary	1.307 (1.047–1.559)	$+0.427$ [$+0.236$, $+0.601$]	0.030	—
ZHIZHI, preregistered	Frozen rules, nominal value (H1)	1.734 (1.372–2.065)	—	—	1.541
ZHIZHI, v6.1 AMBER	Repeatable triggers booked at one half, a quarter under an independent condition (Section III-D)	1.513	—	—	1.400
ZHIZHI, v7 conventions†	v6.1 + the revisions of Section IX + the conventions of Table VII	1.100	—	—	1.274

ΔRMSE = RMSE(preregistered ZHIZHI) − RMSE(method), a negative value favouring ZHIZHI; paired percentile bootstrap over cards, 2000 resamples, seed 20260924; RMSE intervals as for H1; p_{W}: Wilcoxon signed-rank on paired squared errors. B0–B2 are fitted on the 699 units and spells of the four earlier sets and *Proving Grounds*, and under leave-one-out refitted per held-out card on the same 481 cards as ZHIZHI (B2's penalty fixed); B3 was not run on the earlier sets, and 60-card intervals were not computed for the last two rows. ZHIZHI leave-one-out: v6.1 AMBER 1.400 (1.438 with every repeatable trigger at one half); v7 with its parameters re-estimated leave-one-out 1.274 (ΔRMSE against 1.400: -0.127, -0.239 to -0.033). Against B3 (first run) the v6.1 row gives $+0.206$ [$+0.015$, $+0.384$] and against the mean of the five runs (1.217) -0.117 [-0.334, $+0.083$]; the v7 row's comparisons are in the text. On the 54 cards whose rules text was available in Chinese, B3 $+0.467$ [$+0.258$, $+0.667$] and B2 -0.110 [-0.498, $+0.255$]. B3's further runs, MAE, units and spells separately, a gradient-boosting control, B2's feature list and the B3 protocol with its isolation audit: App. S-VI.

classes the frozen rules booked at zero or deferred. Read back into the earlier sets, they share one principle: the designer prices a card for the situation in which it would be played—whatever the card loses if its effect does not fire is priced as if it fired, and whatever costs nothing when unused is given away. Table VII states the nine conventions this yields (evidence card by card in App. S-VII); three are well supported: count-dependent effects are booked at two instances (31 cards, implied count median 2.00 for units and spells alike; a pre-unblinding check on 14 area-damage spells had found 2.0); an optional additional cost buying an on-play effect is not charged for (12 of 13 cards priced exactly at the unpaid version); and symmetric kills, zero in v6.1, are priced at 8 C per enemy unit and -4 C per own unit, two enemy and one own by default (*Cull the Weak* (OGN-209), *Safety Inspector* (UNL-164) and *Unchecked Power* (OGN-123) are then priced exactly, at 4, 7 and 11 C).

With the conventions applied and their parameters re-estimated leave-one-out, the error on the 481 earlier-set cards falls from 1.400 (the AMBER convention) to 1.274 C (ΔRMSE -0.127, 90% interval -0.239 to -0.033; on the 37 cards that change, 2.598 to 1.533 C), and on the 60 primary fifth-set cards from 1.513 to 1.100 C (eleven cards change; 1.181 C with only the count default and the optional cost reduction, both fixed on the earlier sets).

These conventions are exploratory in a stronger sense than the prices above: the classes were sought only after the fifth-set misses had been seen, so what to look for is post hoc even where the parameters come from the earlier sets; several readings were fixed by the first author after seeing the data (own losses at -4 C, which two earlier-set cards with one unit on each side fit exactly; the 2/1 default for Unchecked Power; two card-specific readings named in App. S-VII); conventions 6 and 7 rest on four and two cards; and the exceptions in Table VII are listed, not modelled. Every convention that changes a price improves leave-one-out prediction on the cards it changes; the staged stun price above does not (six earlier-set cards, 0.951 to 1.819 C). The revised model is the most accurate version of the table to date; against B3 it is at least level, not clearly ahead (Section VIII-C), and its confirmation is left to a preregistration on the sixth set.

X. DISCUSSION

What the designer's ledger looks like. Read together, the recovered prices describe a costing logic with a few strong commitments: a fixed 2:1 exchange between power and energy; a linear price for damage and might up to a cap equal to the unconditional-kill price; hand cards as an asset with a reserve price while the act of playing is rent-free; a discount of one half for repeatable and player-controlled contingencies and nothing for a one-off condition on the game state; free play-timing; and integer rounding that gives cheap units their effects nearly for free. These are rules a design team could maintain in a spreadsheet—weak evidence that something real, not an artefact of the fit, has been recovered—and the conventions of Section IX extend the same logic to count-dependent, symmetric and optional effects. The price of scoring one point (11.50 C) is a design-side threshold, not an exchange rate for live play, the marginal value of a point being strongly non-linear.

Transferability. The method needs four things of a game: a cost expressible in one numéraire; effects that card text lets one enumerate into a finite set of types; a vanilla-statistic curve fixing the budget a plain card buys; and enough sets—three or four, several hundred costed cards—for anchor pairs isolating single effects to exist and for a later set to be held out. The protocol needs only a release calendar. What is *Riftbound*-specific is content, not procedure: the two-currency cost and its exchange rate, channelled runes, uncosted battlefields and

TABLE VII
REVERSE-ENGINEERED COSTING CONVENTIONS† (EXPLORATORY: CLASSES IDENTIFIED AFTER THE FIFTH SET WAS UNBLINDED; PARAMETERS FROM THE EARLIER SETS UNDER LEAVE-ONE-OUT). F IS THE VALUE OF THE REST OF THE CARD; PER-CARD EVIDENCE IN APP. S-VII

#	Convention†	v6.1	Evidence (cards; fit)	Confidence
1	Count-dependent effects (bonus per X, cost reduction per X, damage to every enemy unit) booked at two instances. Exceptions listed, not modelled: counts of exhausted units or trash cards, cheap units whose might fills their cost, Avalanche	Zero instances, or deferred (all or indefinite targets)	31 cards: implied count N^* median 2.00 for units and for spells; leave-one-out RMSE on the 24 per-X cards 3.30 C at zero instances, 1.76 at two (lowest); 14 area-damage spells checked before unblinding: 2.0; the 19 earlier-set cards it changes: 3.044 $\rightarrow$ 1.757, 3 exact	High
2	"Up to N" targets booked at $\min(N, 2)$	One target in the base scenario	8 earlier-set cards: leave-one-out RMSE 2.272 $\rightarrow$ 0.948, 3 exact	Medium–high
3	Symmetric kill or damage: enemy losses at full price (kill 8 C, also when the opponent chooses the unit), own losses at one half (kill -4 C, damage halved); default two enemy units, one own	Booked at zero; opponent-chosen kill 1.40 C	7 cards, predicted against printed: 3 exact (Cull the Weak 4, Safety Inspector 7, Unchecked Power 11), 2 within 0.5 C (Tibbers 11.5 / 12, Flurry of Blades 1.5 / 1), King's Edict 8 / 10, Avalanche 4.5 / 9	Own -4: high (two exact cards); 2 / 1 default: medium (fixed after unblinding)
4	Kill a friendly unit as an additional cost: -2 C, distinct from the -4 C of a symmetric kill	Same	Cruel Patron exact; Commander Ledros trades each friendly kill for 2 C of cost	High
5	Optional additional cost buying an on-play effect: the printed cost pays for the unpaid version only	Same base value; the paid version in the full-trigger reference	13 cards: 12 exact (Zed, From the Shadows $+2$); net value of the paid option median 0.00	High
6	Optional cost reduction ("you may pay x as an additional cost; if you do, my cost is reduced by r") assumed taken: predicted cost $F + r$	F	4 cards: RMSE 1.936 at F, 1.250 at $F + r - x$, 0.707 at $F + r$	Medium (4 cards)
7	Optional on-play payment ("when you play me, you may pay x to …") booked at one half of its net value	Zero	2 informative cards: f^* 0.50 and 0.71 (a third has net value 0); RMSE at 0 / 0.5 / 1: 3.000 / 1.041 / 1.414	Low (2 cards)
8	Conditional cost reduction ("if …, my cost is reduced by r"): units not charged; spells unchanged (evidence too thin)	Zero	Units: 4 cards, f^* median 0.00 (3 exact); spells: 5 cards, median $+0.30$, not adopted (booking them at 0.40 moves the fifth-set error from 1.100 to 1.105 C)	Units high; spells not adopted
9	Units of 7 C or more: effect value separated with the might budget $B(c)$ when inverting a convention; value against cost otherwise 1:1	Separation of Section IV-H, not used in the inversions	Count-dependent cards, leave-one-out: 1.698 C at 1:1, 1.631 with the budget on units $\geq$ 7 C (-0.067, -0.278 to $+0.109$); converting every unit through the curve: $+0.142$ over the 481 cards	Medium

legends scored per use, and the keyword set. MTG, *Lorcana* and the *Pokémon* TCG meet the four conditions on paper; the transfer is untested and we claim nothing about it. *Use.* A price table with provenance is a natural constraint for procedural card generation [4], [23] and a simulation-free fitness signal for automatic balancing [15], [18].

A. Limitations

Scope. The model addresses unit fairness only; it ignores synergy (apart from the complementarity premium), tempo and metagame; η is not a strength rating; and the 0.86–1.17 band is a working reference, not a tested one. *Trigger booking.* The 0.5 and 0.25 factors are conventions estimated on the same pool; the 0.5 factor passed its preregistered test on 19 units, narrowly in its median (H9a), the 0.25 factor could not be tested, and tagging a trigger as conditional is a rules-text judgement (17 of 162) made after reviewing the ranked list. *Sample size and coverage.* Most parameters rest on 3–15 cards and the resolution is about 0.3 C (low-confidence prices in App. S-I); gear is priced with an expected-uses parameter the ablation does not test, and 205 units and spells are partial or deferred. *Blind test.* H1 passes on the point estimate, as registered, but its 90% interval reaches above the threshold. *Reproducibility.* After unblinding we re-derived the five most extreme fifth-set spells from card text and the frozen rules, with an independent second check (App. S-V): none was a misreading of card text; three expose gaps in v6.1 that the blind test was designed to surface—a "can't move this turn" clause the rules did not price, and count-scaled spell effects that v6.1 kept at zero while pricing the same mechanism on units at 0.5 activations; one card is simply strong and one is champion-bound. The next version patches all three gaps; the preregistered numbers are left untouched. *Researcher degrees of freedom.* The rule structure (effect types, deferral rules, factors) was fixed by hand on the four calibration sets, and leave-one-out removes only the leakage of parameter estimates, not of that structure. The primary blind sample is 60 cards, the new-keyword cards were excluded as partial, and the fully priced share fell from 71% to 54%. The blind pricer's isolation from the costs rested on instructions, not a technical barrier; its instructions and process record are published. The four pre-unblinding revisions were the first author's; the original file is published beside the revised one. The new-keyword and stun prices and the conventions of Section IX were estimated after the costs were read, carry the further limitations stated there, and await a preregistered test on the sixth set.

XI. CONCLUSION

A CCG designer's internal costing model can be recovered from printed cards alone by treating pricing as an inverse hedonic problem: ZHIZHI inverts effect prices from zero-residual anchor pairs and imputes every card's value from its text without refitting, and the AMBER score reports the result per card. For Riftbound the recovered table reproduces printed costs for the central mass of 497 cards (median AMBER score 1.00, IQR 0.86–1.07), predicts held-out cards and exposes unstated design rules. A blind test registered before the next set's preview season confirms the central claim out of sample: the frozen table priced the fifth set with RMSE 1.73 C, inside the registered threshold of 1.84 C, at a median nominal score of exactly 1.00, and the half-activation booking lowered the RMSE on its 19 trigger units from 2.00 to 1.30 C. Its misses,

read back into the earlier sets, yield nine costing conventions that, in a revision made after unblinding, lower the error to 1.27 C on held-out earlier cards and 1.10 C on the new set—the most accurate version of the table to date. A language model guessing from card text beats the frozen table and is level with the revised one; what the table adds is a price for every effect that can be read off, traced to anchor cards and registered before the next set is revealed. The procedure is the general result: invert prices from anchor cards, freeze and register the table, price the next set blind, and read the misses back as conventions to be registered in turn; its next iteration here registers the revised table before the sixth set.

Data Availability

The card export, per-card valuations and scores, the anchor table and the audit change log will be released on Zenodo. The blind test is preregistered at https://doi.org/10.5281/zenodo.22928382 (H1–H8, 24 September 2026; H9–H10 added as version 2, https://doi.org/10.5281/zenodo.22950713, the same day, before any *Radiance* card was priced); the blind pricing, original and revised, with revisions, protocol, subagent instructions and SHA-256 hashes, is version 3 (https://doi.org/10.5281/zenodo.23306713, 12 October), published before the costs were read. The unblinding report, scripts and post-hoc analyses will accompany the dataset.

Acknowledgment

Two AI systems were used. Google Antigravity exported the card data from Rift Atlas and parsed rules text into effect annotations for early versions of the value table. Claude (Anthropic) verified the authors' pricing calculations, implemented and ran the robustness analyses, typeset the paper and drafted and edited parts of the English text; a separate Claude session priced the *Radiance* blind table under the frozen rules without access to the costs (Section VIII-A); Claude also wrote and ran the unblinding scripts and the post-hoc analyses of Sections VIII and IX (baselines and conventions), which the authors reviewed and in part revised (the pre-unblinding corrections and the readings named in Section IX are the first author's); baseline B3 is a set of separate Claude sessions given the blind table only, and a further session without access to the analysis checked the decompositions behind the conventions. The pricing calculations and the structural analysis of the card pool are the authors' own, who reviewed every AI-assisted step and take full responsibility for the content.

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APPENDICES

Appendices S-I to S-VII give in full the price tables, identifications, analyses and per-card evidence that the main text summarises; the main text refers to them as App. S-I to S-VII. Tables and figures in the appendices are numbered S1, S2, ...

APPENDIX S-I. PRICE TABLES IN FULL

These are the price tables in full: the base prices by confidence tier, the selection effects priced by the order-statistics model, the discount factors for target restrictions, the prices for extended mechanisms, and the low-confidence prices with their treatment. The main text reproduces ten rows of the first in its Table II. The identification behind each row is in App. S-II.

APPENDIX S-II. IDENTIFICATION DETAILS AND PER-CARD EVIDENCE

This appendix gives the identification of each mechanism price that the main text summarises in Section IV-B and Table II: the budget-slope intervals, the sensitivity of the DEATHKNELL coefficient to the separation convention, and the subsections on PREDICT, modal effects, FLOW, tokens, DEATHKNELL, target restrictions, on-play effects, the extended mechanisms and the same-target complementarity premium. Table numbers refer to the price tables of App. S-I; section and equation numbers to the main text.

S-II.A The might budget: identification intervals

Vanilla units (no effect text) satisfy might = cost up to 5 C, after which might grows faster than cost: *Mega-Mech* (OGN-088) is 7 C for 8 might and *Mountain Drake* (OGN-142) is 9 C for 10 might. Every vanilla unit satisfies might $= \lfloor B(c) \rfloor$ for the budget function

$$B(c) = \begin{cases} 1.1\,c, & c \le 5, \\ 5.5 + 1.33\,(c-5), & c > 5, \end{cases} \tag{5}$$

measured in C, since one might is defined to be worth 1 C. Under the floor convention the two slopes are identified only up to intervals: every slope in $[1.00, 1.20)$ reproduces the five vanilla units at 1–5 C, and every slope in $[1.25, 1.375)$ reproduces Mega-Mech and Mountain Drake; 1.1 and 1.33 are the values used in the value table. The rounded budget $\lfloor B(c) \rfloor$ is the same for every slope in these intervals at 1–9 C, so the might budgets, the DEATHKNELL fit and the separation of effect value below do not depend on the choice. The unrounded slack $B(c) - c$ does (at 4 C it ranges from 0 to 0.8 C over the first interval), and the three places that use the unrounded value—the on-play multiplier of Section IV-H, the keyword-rounding boundary of Section VI-C and the might-only sensitivity check of Section IV-H—are marked as such. $B(c)$ is the total value the designer expects a c-cost unit to deliver. It is used (i) as a reference ratio $\eta_B = V/B(c)$ and (ii) to separate effect value on expensive units. The η denominator remains C_{tot}.

S-II.B Separating effect value from might

For spells, effect value is C_{tot} minus all other priced components. For units above 5 C, effect value is $B(c) - m -$ (other priced components), all in C; the implicit assumption is that the high-cost scale discount applies to the whole value budget rather than to might alone. This is supported by data: 19 units carrying a single on-play effect, separated in this way, yield effect values equal to $1.10\times$ (resource effects) and $1.12\times$ (removal effects) the price of the same effect on a spell, i.e. approximately full price. The excess over 1.00 is the fractional budget slack: under whole-card rounding $V = \lfloor B(c) \rfloor$, the six such units whose effect is integer-valued (*Stormclaw Ursine* (OGN-137), *Riptide Rex* (OGN-092), *Harnessed Dragon* (OGN-234), *Beast Below* (SFD-132), *Cemetery Attendant* (OGN-165), *Ocean Drake* (VEN-115)) match exactly, so the multiplier is kept at 1.00. Under the alternative (discount applies to might only, so surrendered might is converted at the marginal rate $1/1.33 \approx 0.75$ C per might), the same effects would be worth only $\approx 0.8\times$ spell price. For units at or below 5 C the sub-1 C budget slack is rounding and is not credited as effect value; effect value is simply $c - m$.

Sensitivity. The DEATHKNELL coefficient λ (the fraction of a Deathknell effect's unconditional value that is booked; App. S-II.C, Deathknell) is 0.50 (median) and 0.46 (least squares) under the whole-budget convention. Under the might-only convention (which converts the unrounded shortfall $B(c) - m$ at 0.75 C per might, whereas the adopted fit uses $\lfloor B(c) \rfloor$), *Ruined Rex* (UNL-067) moves from 0.75 to 0.66 and *Tasty Faefolk* (OGN-075) from 0.33 to 0.27, leaving the five-card median at 0.50 and least squares at 0.41; the conclusion $\lambda \approx 0.5$ is insensitive to the convention.

S-II.C Mechanism prices in full

Predict and Look-N-Take-1: PREDICT N does not change hand size; it improves the quality of subsequent draws. The order-statistics model prices it with random-card quality fluctuating around the draw value of 2 C, with dispersion calibrated from the cost gap between *Clairvoyance* (VEN-056) (7 C, Predict 5 then draw 2) and *Consult the Past* (OGN-083) (4 C, draw 2). Jointly fitting Clairvoyance, *Apprentice Mage* (VEN-047) and the discard price gives 0.385 C per card (least squares) and 0.40 C (minimax); we adopt 0.40 C, and uniform versus normal noise changes the result by at most 0.05 C. Table S2 gives the derived values. VISION is Predict 1. A unit carrying Vision is credited the full 0.40 C at 4 C and above, and 0.12 C on a 2–3 C unit under keyword rounding (Section VI-C).

Look-3-take-1 can be read as "draw 3, then discard the worst two of the three new cards": under the same dispersion the three order statistics of three fresh draws sum to 6.00 C, so the reading is an identity rather than a check (the expected values of the two discards, 1.17 and 2.00 C, leave exactly the 2.83 C of Table S2). The discard price of -1.00 C is the minimum of a whole hand, not of three fresh draws; the two follow from the same model and are not in tension.

Modal "Choose One" Effects: A modal card is worth its best single mode plus a selection premium. Calibrating $\sigma \approx 1.4$ C on

TABLE S1
BASE PRICES RECOVERED BY ZHIZHI (STAGE 1), IN C, GROUPED BY CONFIDENCE TIER. H = TWO OR MORE INDEPENDENT CARDS AGREE; M = SINGLE METHOD OR FEW SAMPLES; M–L = SINGLE ANCHOR CARD; L = ASSUMPTION-BASED. †REVISED AFTER THE FIFTH SET (SECTION IX); ALL STATISTICS OUTSIDE SECTION IX USE THE FROZEN PRICE.

Effect	Price	Conf.	Identifying evidence
1 power (rune icon)	2.00	H	Hextech Ray, Falling Star, Punch First (exact); closes with rune channelling below
1 might	1.00	H	5 vanilla units; 18 further units with only zero-priced keywords
1 damage	1.00	H	Incinerate vs. Final Spark; no premium for battlefield-only targets
$+1$ / -1 might this turn	1.00 / 1.00	H	Guttural Roar (VEN-072), Punch First; Frigid Touch (SFD-066), Smoke Screen (OGN-093); both realised only through combat, $\times 0.50$ when usable only in the main phase (Table S4)
SHIELD X, ASSAULT X	0.50 per point	H	Half price (conditional on attacking/defending); same at $X = 1$ and when granted; 0.15 when native to a 2–3 C unit (keyword rounding, Section VI-C)
Draw 1	2.00	H	Progress Day (OGN-114): 8 C for draw 4; Dredge Up, Downstage Dramatics, Consult the Past
Channel 1 exhausted rune	2.00	H	Mobilize (OGN-134), Catalyst of Aeons (OGN-138); Qiyana, Victorious "draw 1 or channel 1"
Gain 1 power	2.00	H	Same as paying it (no discount for restricted use or timing); Seal of Rage (OGN-040) and five other Seals, Honeyfruit (UNL-049)
1 Gold gear token	0.875	H	Trove Golem (SFD-174) 0.75 and Deadly Flourish (UNL-073) 1.00, midpoint
Discard 1 (own hand)	-1.00	H	Lunar Boon (UNL-125), Chemtech Enforcer (OGN-003); lowest-quality card in hand
ACTION, REACTION, REPEAT, HIDDEN, ACCELERATE, DEFLECT, WEAPONMASTER, AMBUSH, TANK	0	H	Multiple cards each; timing analysis in Section VI-B. The keywords themselves are free; a paid Accelerate prices "enters ready" at 2.00, the same as readying a unit
GANKING	1.00	H	Kinkou Lifeblade (VEN-093), Brutal Hunter (VEN-070) gain Ganking on Empower at exactly 1.00; innate mean $\approx$1.1
Move 1 enemy unit (any destination)	3.00	H	Charm (OGN-043); Skyward Strike (UNL-038), Void Assault (UNL-202), Moonfall (UNL-198)
Move 1 friendly unit	1.00	H	Flash (OGS-011), Void Assault (UNL-202), Shuriken Flip (VEN-140)
Ready a unit	2.00	H	Upstage Comedy (UNL-009), Wallop (OGN-146); Meditation (OGN-048) in reverse
Kill a unit (unrestricted)	8.00	H	Vengeance (OGN-229), Blast of Power (OGS-012); Hidden Blade (OGN-213) consistent
Kill a friendly unit (as additional cost)	-2.00	H	Cruel Patron (OGN-208), exact; Commander Ledros (OGN-231): each friendly unit killed takes one power (2.00) off its cost, net 0. Sacrifice (UNL-173) asks for a Mighty ally (≥ 5 might), a dearer cost (Table S5)
Symmetric kill: each enemy unit / each own unit† (exploratory; not in the frozen table)	8.00 / -4.00	H	Cull the Weak (OGN-209), Safety Inspector (UNL-164): one unit each side, both exact (Convention 3)
Return enemy / own unit to hand	6.00 / -1.00	H	Rebuke (OGN-172); Retreat (OGN-104), Star-Crossed (UNL-128)
Score 1 point	11.50	H	6 cards, mean = median
May be played to an open (or attacking) battlefield	$+1.00$	H	Sneaky Deckhand (OGN-176), Rengar, Pouncing (SFD-025), Ocean Drake (VEN-115) at 1.00; Sai Scout (OGN-174) 0.60
Remove a buff (additional cost)	-1.00	H	Symmetric with buff = 1.00
Stun an enemy unit (any / restricted)	4.00 / 2.00	M–L / H	Any: Rune Prison (OGN-050), single anchor; restricted: Facebreaker (OGN-220; the ally it also stuns at 0), Thwonk! (SFD-040), Heroic Charge (UNL-155)
Stun an enemy unit, from the second set† (staged reading, exploratory; not in the frozen table)	2.00	H	Ki Burst (RAD-028), One-Two Punch (RAD-121); Back Off (UNL-042) with its self-controlled draw at one half; signature spells (Zenith Blade) not used (Section IX)
Survive: remove damage, exhaust and recall a unit	2.00	M	Zhonya's Hourglass (OGN-077, anchor), Tactical Retreat (UNL-175); Guardian Angel (SFD-051) consistent
Kill a gear (unrestricted)	2.75	M	Median of 4 unconditional cards (Section VI-D)
Gain 1 XP / spend 1 XP	0.40 / 0.50	M	18 XP-gaining units; XP-to-buff conversion (Section VI-D)
Ready a rune / gain N energy	1.00 / N	M	One rune = one energy
BACKLINE	0	M	As TANK, DEFLECT
Opponent discards 1 (their choice)	1.00	M–L	Bewitching Spirit (UNL-121)
Return a unit from trash to hand	2.00	M–L	Morbid Return (OGN-170)
May be played to an occupied enemy battlefield	$+2.00$	M–L	Dauntless Vanguard (SFD-093); Deadbloom Predator (OGN-161) lies beyond the budget curve and is not used
Return or kill own gear (additional cost)	-1.00	M–L	Legion Quartermaster (SFD-044)
Opponent kills one of their own units (their choice)	1.40	L	Opponent picks the weakest; booked as one 1-might token (revised: 8.00 in a symmetric kill†, above)
Can't be chosen by enemy spells/abilities	unpriced	—	High practical ceiling; no reliable anchor; cards carrying it are marked partial

Qiyana, Victorious (OGN-155) ($+0.85$ C), *Buhru Captain* (SFD-091) ($+0.50$ C) and *Disposal Order* (UNL-103) (0) gives, for

TABLE S2
SELECTION EFFECTS PRICED BY THE ORDER-STATISTICS MODEL

Effect	Value (C)
PREDICT N (VISION = Predict 1)	$0.40\,N$
Look 2, take 1 / look 3, take 1	2.55 / 2.83
Look 4, take 1 / look 5, take 1	≈3.00 / 3.12

TABLE S3
DISCOUNT FACTORS FOR TARGET RESTRICTIONS (ZHIZHI STAGE 1)

Restriction	Factor	Evidence
At a battlefield only	1.00	Incinerate, Blast of Power
State/position (attacker, same place, pre-damaged)	0.50	Thwonk!, Heroic Charge, Facebreaker (OGN-220)
Destination restricted	0.67	Isolate (UNL-124), Temptation (SFD-129)
Kill, might ≤ 3	0.55	Soul Harvest, Lacerate (VEN-127), Sandshifter (SFD-158)
Return to hand, might ≤ 3	0.20	Gust (OGN-169), Windsinger (SFD-138)
Kill, might ≤ 2	0.15	Blood Money (single anchor; net of its Gold token)
Trait or relative-might restriction	0.50	Decree of Unity (VEN-131), Public Execution (VEN-154)

equally valued modes, $+0.8$ C (two modes), $+1.2$ C (three) and $\approx +1.45$ C (four): the increment falls from ≈0.4 C to ≈0.25 C as modes are added. The premium shrinks with the gap between modes: $+0.4$ C at a 1 C gap and $+0.16$ C at 2 C. When the *opponent* chooses, the sign reverses: *Shakedown* (OGN-033) (modes worth 4 C and 6 C) is worth $4 - 0.16 = 3.84$ C against a printed 4 C.

Flow: A Flow card is played twice. The first play, from hand, pays the printed cost and consumes a hand card; the value k of the Flow keyword is booked here. The Flow play pays the Flow cost F and consumes no hand card, yielding the effect value V plus the saved hand card ($h = 1.00$ C, Section IV-D). On the printed side,

$$k = 0.2\,\max\big(0,\,1-p\big), \qquad p = F - C_{\mathrm{tot}}. \tag{6}$$

The coefficient is estimated from the five spells with $p = 0$, where implied $k = C_{\mathrm{tot}} - V$: *Dredge Up* (VEN-049) (2 C, draw 1) gives 0; *Up from the Deep* (VEN-100) (3 C, two 1-might Tentacle tokens at 2.8 C) 0.2; *Iterative Design* (VEN-051) (4 C, a 3-might Mech token at 3.4 C) 0.6; *Onslaught* (VEN-081) (4 C, $+6$ might at the non-showdown discount, 3 C) 1.0; *Perfect Execution* (VEN-012) (5 C, effect 4.5 C including the same-target complementarity premium of Section V-C) 0.5. The median of the five is 0.5 C, the mean 0.46 C. However Onslaught's effect value depends on the non-showdown discount, which is itself supported mainly by Onslaught; using it to raise k would be circular, and without it the median of the remaining four is 0.35 C (mean 0.33 C).

The parameter cannot be pinned down precisely. The seven scored Flow cards with $p \geq 1$ (an eighth, *Twilight Shroud* (VEN-031), is unscored because it grants "can't be chosen") should show zero Flow premium in theory but range from -1.8 to $+2.5$ C (s.d. ≈1.4 C), which is the noise of the effect prices themselves; with that noise the four-card standard error is ≈0.7 C, and ± 0.25 C precision would need about 31 cards with $p = 0$. The $p = 0$ and $p \geq 1$ groups are statistically indistinguishable (exact Mann–Whitney $p = 0.52$). Independent evidence comes from REPEAT, which is also "repeat the effect without spending a hand card": of seven Repeat spells whose effect is priced by other cards (Downstage Dramatics, UNL-061; Upstage Comedy, UNL-009; Feral Strength, SFD-034; Frigid Touch, SFD-066; Bonds of Strength, SFD-151; Blood Rush, SFD-003; Desert's Call, SFD-031), six carry a Repeat premium of exactly 0 and one of -0.4 C. Since Flow can be deferred to later turns it should be worth at least as much as Repeat. The plausible range is thus 0–0.6 C; we adopt the midpoint of the direct estimate (0.35 C without Onslaught) and the Repeat evidence (0), rounded to 0.2 C, and list it among low-confidence parameters. The parameter review of Section VII-C quantifies the uncertainty: on the four direct cards the exact bootstrap 90% interval of the midpoint rule is $[0.00, 0.28]$ (unshrunk median $[0.00, 0.55]$); fixing k at any value from 0 to 0.6 moves the full model's leave-one-out RMSE only in the fourth decimal; eight estimation rules, $k \equiv 0$ among them, have indistinguishable leave-one-out error on the four cards; and a joint fit of k and the truncation point over the eleven scored Flow cards other than Onslaught cannot identify the truncation. *Shuriken Flip* (VEN-140) with $p \geq 1$ recovers 0, consistent with the truncation in (6).

Each Flow card is reported under four regimes: printed $\eta = (V+k)/C_{\mathrm{tot}}$; first play only V/C_{tot}; Flow play $(V+1)/F$; both plays $(2V+1)/(C_{\mathrm{tot}}+F)$. When $F = C_{\mathrm{tot}}$ the Flow play is the more resource-efficient (Dredge Up: 1.00 first, 1.50 on Flow); at $F = C_{\mathrm{tot}} + 1$ the two are roughly equal; at $F \geq C_{\mathrm{tot}} + 2$ the Flow play is less efficient.

Unit Tokens: A token unit is worth might $+0.40$ C: 13 usable samples (spells, on-play effects and Deathknell) have median $+0.40$ C, unchanged when Deathknell samples are removed. Within-type dispersion (Recruit tokens $+0.25$ to $+1.55$ C) exceeds between-type differences, so one formula is used. Mechanically special tokens are priced individually: the Shadow Clone token ≈3.9 C; a 3-might TEMPORARY Sprite entering ready ≈2.75 C.

Deathknell: A DEATHKNELL effect is booked at $\lambda \approx 0.50$ of its unconditional value. If it were booked in full, Deathknell units would systematically exceed their cost; the fit therefore solves for λ such that might $+\lambda\times$(unconditional value) equals the budget might $\lfloor B(c) \rfloor$, i.e. $\lambda = (\lfloor B(c) \rfloor - m)/V_{\mathrm{DK}}$, where V_{DK} is the unconditional value of the Deathknell effect. For the single-Deathknell units: *Watchful Sentry* (OGN-096) $(2-1)/2 = 0.50$; *Soaring Scout* (OGN-216) 0.50; *Black Rose Dignitary* (UNL-152) $(3-2-0.15)/2 = 0.43$ (Assault 1 at 0.15 under keyword rounding, Section VI-C); *Ruined Rex* (UNL-067) $(9-6)/4 = 0.75$; *Tasty Faefolk* (OGN-075) $(8-6)/6 = 0.33$; *Undercover Agent* (OGN-178) $(8-5)/2 = 1.50$. Excluding the outlier Undercover Agent (a first-wave card), the five remaining samples give: least squares through the origin 0.46, minimax 0.54, median 0.50, mean 0.50 (t-based 95% CI 0.31–0.69). All four statistics fall in 0.46–0.54; we adopt $\lambda = 0.50$. *Honest*

TABLE S4
PRICES FOR EXTENDED MECHANISMS RECOVERED BY ZHIZHI (STAGE 1), IN C (†ADDED AFTER THE FIFTH SET, SECTION IX)

Mechanism	Price	Evidence
Play-timing flexibility (ACTION, REACTION, HIDDEN, REPEAT)	0	Group comparison over 47 spells (Section VI-B)
On-play effect on a unit (resource, removal, other persistent effects)	spell price $\times 1.00$	Resource effects: 7 cards, median 1.10; removal: 5 cards, median 1.12; both ratios come from the unrounded $B(c)$; under $\lfloor B(c) \rfloor$ six integer-effect units match exactly
On-play effect realised through same-turn combat (combat trick, incl. $-X$ might this turn)	spell price $\times 0.50$	Regression -0.27 ($p = 0.057$), multiplicative fit 0.55; optional-cost anchors ≥ 0.5; range 0.5–0.75
On-play effect on a 2–3 C unit (included in printed cost)	spell price $\times 0.30$, multiplied with the above	10 cards; regression -0.63 ($p < 0.001$); multiplicative fit 0.26 with combat trick fixed at 0.5
Keywords native to a 2–3 C unit (keyword rounding)	total $<$ 1 C: $\times 0.30$ (VISION 0.12, ASSAULT 1 / SHIELD 1 0.15); $\geq$ 1 C: full	9 units with $<$ 1 C keywords give up no might for the keyword; 5 of 6 with $\geq$ 1 C keywords do (Section VI-C)
EMPOWER X (self-paid)	option 0; bonus at 1:1	Kinkou Lifeblade, Brutal Hunter, Steel Paws (VEN-043), Punching Poro (VEN-007) all 1.00
One free Empower granted by another effect	≈ 2.4	Median Empower cost 4 C $\times$ scenario factor ≈ 0.6; Wild Claw (VEN-089) recovers 2.36 independently
Look at opponent's hand and choose a card to discard/recycle	any 4.00; non-unit 3.00; single domain 1.00	Mindsplitter (OGN-192, 11 C, 7 might; budget-method ceiling ≈ 6.5), Sabotage (OGN-156), Decree of Strength (VEN-085)
Gain XP	0.40 per point	18 units; through-origin LS 0.38 with $\lfloor B(c) \rfloor$ (0.59 unrounded); hidden by rounding on cheap units
Spend XP	0.50 per point	2 XP buys one buff ($+1$ might)
BURN	0	Blade Twirler (VEN-002), Shadow Order Disciple (VEN-095); effects cancel
Exhaust own unit (additional cost)	-1.00	Opportunity cost: the unit forgoes one move this turn; moving a friendly unit $=$ 1 C; on the fifth set Ki Burst (RAD-028, exhaust an ally to stun) is priced exactly at $2 - 1$
Own unit can't move this turn	$-2.00^{\dagger}$	Not an exhaust: Entrenched Boar (RAD-026), ready but unable to move, priced exactly (net 0); Hunker Down (RAD-034) $+1$ C, as other 1 C cards (Section IX)
Same-unit combination: reposition (move, ready, grant GANKING) and boost (this-turn might, buff, ASSAULT/SHIELD) one friendly unit in one spell	sum of parts $+1.00$	Ride the Wind, Resonating Strike, Showstopper, Vault Breaker each $+1.00$; Perfect Execution $+1.30$; different-target combinations 0 (Section V-C)
Overflow cap for single-target damage / $-X$ might	8.00	Equals unconditional kill; largest linear anchor (Final Spark); only 9 units (1.9%) have might ≥ 9
Spell without ACTION/REACTION/HIDDEN: $\pm X$ might this turn, stun	spell price $\times 0.50$	Same reasoning as combat-trick on-play effects; Onslaught (VEN-081) 4 C for $+6$ (≈ 0.67/pt) vs. Punch First 5 C for $+5$ with Action (1/pt); $-X$ might this turn takes the same factor; ready is not discounted; Assault/Shield grants, already at 0.5 per point, are not discounted again

Broker (SFD-155) serves as a check: 2 C for 2 might plus a Gold token (0.875 C); at $\lambda = 0.5$ the value is $2 + 0.44 \approx 2.44$ C, which rounds to the printed 2 C. Tokens created by Deathknell are discounted at the same rate.

Target-Restriction Discounts: Restricted versions of an effect are priced as a fraction of the unrestricted price (Table S3). Restricted kill is priced *above* equivalent damage: *Soul Harvest* (UNL-159) (4 C, kill a unit with might ≤ 3 at a battlefield) is printed at 4 C, whereas 3 damage costs 3 C, even though 3 damage dominates in use (it also kills $\leq$ 3-might units and combines with combat damage). The model records the designer's price, so the nominal 4 C is retained; in expectation such cards should underperform damage spells of equal cost, an instance of nominal η exceeding effective η. *Blood Money* (SFD-162) (2 C: kill a unit with might ≤ 2 at a battlefield, plus a Gold token) shows no such premium: net of the token the kill is worth about 1.1 C, roughly the damage needed.

On-Play Effects: Cost-Controlled Regression: Units with "When you play me" (on-play) effects require two adjustments. First, effects that must be realised through combat in the same turn—$+X$ might this turn, stun, readying *other* units, granting a this-turn keyword—are worth less on a unit than on a spell: a spell with ACTION or REACTION can be played inside a showdown and realise its effect immediately, whereas a unit can be played only on its owner's turn outside showdowns, before combat, when it is unknown whether combat will occur; and a unit enters exhausted, so a this-turn bonus on itself is nearly unusable. Damage is *not* discounted: its value is settled at resolution (kill or waste) exactly as on a spell, and that risk is already priced into the spell anchors. "$-X$ might this turn" is discounted: it cannot kill by itself (a 0-might unit still needs 1 damage), so, like $+X$ might, it is realised only through combat and takes the same factor. Second, cheap units price on-play effects at a small fraction regardless of type (Section VI-C).

The two effects are confounded: seven of the nine initial combat-trick anchors cost ≤ 4 C. We therefore regress, over 25 units with a single on-play effect priced independently by other cards, the recovered fraction of spell price on two indicators, low cost (2–3 C) and combat-trick type. The low-cost effect is -0.63 ($p < 0.001$) and the combat-trick effect -0.27 ($p = 0.057$); the best multiplicative fit is $0.25\times$ (low cost) and $0.55\times$ (combat trick). Optional-additional-cost anchors, in which the designer prices the effect directly and which are immune to

TABLE S5
LOW-CONFIDENCE PRICES AND THEIR TREATMENT

Item	Treatment	Reason
TEMPORARY	not priced	≈ -2.65 C from two Sprite cards only
Look at opponent's hand, discard any card	4 C	Mindsplitter: 4 C by cost-minus-might, 6.5 C by budget method
Move an enemy unit to its base	2 C	Isolate single anchor; Maddened Marauder (OGN-191) recovers 1 C
Prevent damage	2 C / 4 C	One anchor each
Tokens from Deathknell	token $\times 0.5$	3 samples, mean 0.70 (Machine Evangel 0.95, Ferrous Forerunner 0.44, Carrion Dredger 0.71), dispersed
Flow keyword at $p = 0$	0.2 C (0–0.6)	4 direct cards; per-card noise ± 1.4 C; bootstrap 90% interval of the adopted rule 0.00–0.28
Destroy a gear	2.75 C	Four anchors bimodal (Brittle Steel 4.0, Disarming Rake 3.5, Salvage 2.0, Turn to Dust 2.0); median bootstrap interval 2.0–4.0
Sacrifice a Mighty ally (additional cost)	-5 C (Sacrifice only)	The general price, -2 C, is high-confidence (Cruel Patron, Commander Ledros); Sacrifice alone asks for an ally of 5 might or more and implies -5
Might budget at 7–8 C	$B(c)$ kept	Above 5 C the curve rests on Mega-Mech and Mountain Drake; no vanilla 8 C unit; four fully priced 8 C units total 6.5–8 C against $\lfloor B(8) \rfloor = 9$
Budget-curve slopes 1.1 / 1.33	kept	Identified only to $[1.00, 1.20)$ and $[1.25, 1.375)$ under the floor convention; $\lfloor B(c) \rfloor$ unchanged at 1–9 C; the unrounded slack (keyword-rounding boundary, $1.10\times$ multiplier) depends on the choice
Spell w/o timing keyword: $\pm X$ might this turn, stun	$\times 0.50$	Onslaught is the only usable sample; $-X$ might is included by rule, and its printed anchors (Frostcoat Cub, Blastcone Fae) point to full price

might rounding, give 0.50 for *Masa, Crashing Thunder* (VEN-120), *Nami, Headstrong* (UNL-052) and *Kennen, Keeper of Balance* (VEN-135), with *Gust Monk* (VEN-101) at 1.00 as a single exception; because an optional mode is usually left with some surplus for the player, these values are lower bounds. Among expensive units *Whiteflame Protector* (OGN-082) gives 0.75. Taken together the combat-trick factor lies in 0.50–0.75, and the regression rejects 1.00 for the class as a whole; we adopt 0.50, consistent with the 50% rule of Section VI-A. (Sample sizes differ across the checks in this paper—19, 25, and $7 + 5$ units—because each uses the units that satisfy its own inclusion rule: single effect, effect priced independently elsewhere, effect type, and cost band.) Units with HIDDEN can be played as reactions but only where the player already has a unit, so they take the same factor (*Teemo, Scout* (OGN-197) recovers 0.33). *Frostcoat Cub* (SFD-067) (optional cost) and *Blastcone Fae* (OGN-097) (HIDDEN, on-play -2) are printed at the full $-X$ price, i.e. the designer then priced $-X$ might like damage; under the model's rule both come out conservative (full-trigger η 0.80 and 0.75). Neither is in the regression. Units that enter ready (priced by the ACCELERATE anchor), permanent effects (buffs, kills, moves) and already half-priced SHIELD/ASSAULT grants are not discounted again.

After both adjustments, fully priced units with $\eta \geq 1.5$ fell from 8 to 1 (the survivor, *Ol' Poro* (VEN-029), was later moved to partial status because "cannot be played on your first three turns" is a unique mechanic) and units with $\eta > 1.17$ from 29 to 13 (17 before keyword rounding, Section VI-C), while the median and quartiles (1.00; 0.75–1.00) were unchanged.

Other Mechanisms: Table S4 collects the remaining priced mechanisms. Repeatable effects ("whenever," "when I move," activated abilities, auras) are reported in 0–3 bins with the single-trigger value taken from the price table; an activated ability's per-use net value is its effect value minus its activation cost (energy 1 C, power 2 C, and -1 C if the cost includes exhausting the unit itself).

Same-Target Complementarity Premium: The additive core prices a spell as the sum of its parts. One class of spells departs from this in a consistent direction. Among the eight fully priced spells in which the price of "move a friendly unit" is the only unknown, the implied price of the move splits into two clusters: four spells whose other effect targets a different unit, or that have no other effect, all give exactly 1.00 C (*Flash* (OGS-011), *Shuriken Flip* (VEN-140), *Void Assault* (UNL-202), *Tactical Retreat* (UNL-175)); three spells that first reposition a friendly unit and then boost the *same* unit all give exactly 2.00 C (*Ride the Wind* (OGN-173), move then ready; *Showstopper* (OGN-270), buff then move; *Resonating Strike* (VEN-034), move then $+2$ might). No single value of the move price fits both clusters, whereas the structural split fits seven of the eight cards exactly (the remaining card, *Zenith Blade* (OGN-262)—7 C to stun an enemy and optionally move a friendly unit to it—implies 3.00 C for the move, 2 C above the different-target cluster; it is the card behind the "≥ 6 C" bound on the unrestricted stun in Table S5). The same pattern appears in *Vault Breaker* (UNL-010) (grant ASSAULT 2 and GANKING to one unit: 2.00 C of parts for 3 C) and *Perfect Execution* (VEN-012) (ready a unit and grant it ASSAULT 3: 3.5 C of parts plus the Flow keyword for 5 C). All five cards carry a premium of 1.0–1.3 C over their additive value, and no spell of this shape is priced additively; spells whose positioning and boost act on different units (Shuriken Flip, Void Assault, *Heroic Charge* (UNL-155), *Moonfall* (UNL-198)) have zero residual.

We therefore book a *same-target complementarity premium* of $+1.00$ C for a spell that both positions a friendly unit (move, ready, or grant GANKING) and enhances the same unit for combat ($\pm X$ might this turn, a buff, ASSAULT/SHIELD). Four of the five carriers sit at exactly $+1.00$ and one (Perfect Execution) at $+1.30$; no spell in the pool contradicts the rule, whereas the four spells that split positioning and enhancement across different units show zero residual. Because those four also resolve as a single spell, the premium tracks target identity rather than single resolution: it is an interaction term in

an otherwise additive hedonic model. A natural reading is complementarity—an enhancement placed on the unit that the same card sends into combat is guaranteed to reach the fight it creates. An alternative "atomicity" reading, that one card denies the opponent a response window between the two effects, is not supported: it would predict the same premium for the split-target spells. It is the only interaction term in the model; it applies to spell base values only, since the analogous combinations inside unit triggers and optional modes (e.g. *Pyke, Dockside Butcher* (UNL-028), *Hungry Wolf* (VEN-125)) have no independent cost evidence. With the premium booked, the five cards move from $\eta = 0.67$–0.75 to 0.94–1.00, and Perfect Execution's implied Flow premium falls from 1.5 to 0.5 C (App. S-II.C, Flow). With five supporting cards we treat the rule as a hypothesis. A pattern of the opposite sign—a "then draw a card" rider priced at about half of a stand-alone draw on six spells—was noted in the same review but is left unbooked pending more cards.

Appendix S-III. Additional Tables and the Subsections They Belong To

This appendix holds the full coverage table (Table S6), the subsections that the main text condenses into Sections VI-B–VI-D (timing keywords, cheap units and keyword rounding, selection premia, XP, Flow, scenario effects, Deathknell by set, bans) with their tables, the full ablation and paired-test tables behind the merged Table IV of the main text with their discussion, the parameter review, and the fifteen highest AMBER scores. Section numbers in the text refer to the main text where the section still exists there and to this appendix otherwise.

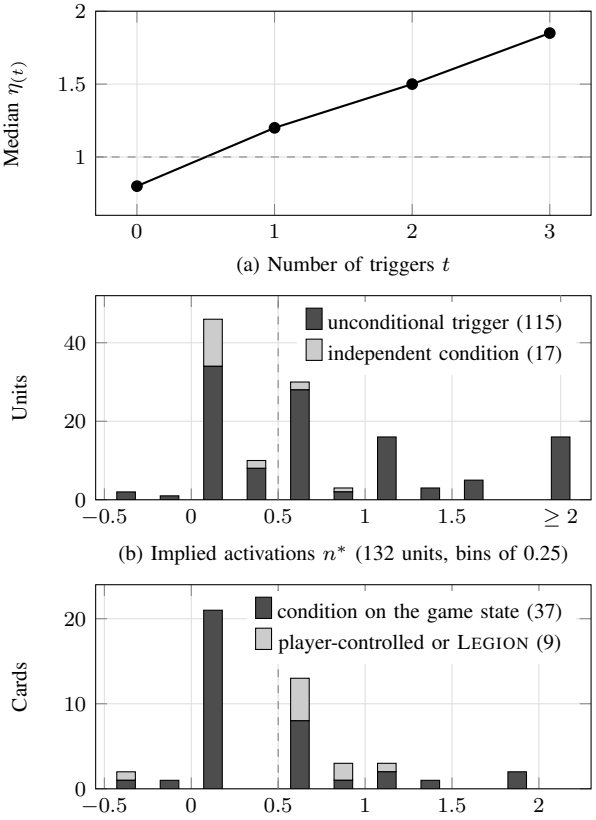
S-III.A Coverage in full

Table S6 gives the coverage by card class with the deferral rules and their counts, and the composition of the fully priced, partially priced and deferred units and spells. Symmetric effects are booked at zero, and a card consisting only of symmetric effects is deferred. All units and spells were re-audited under the final rule set: 88 cards had a value or status revised (mainly 0–3 trigger bins for repeatable effects, full-trigger accounting of conditional effects, optional additional costs booked on the cost side, and reclassification of untargetable or all-target effects), and 28 previously unpriced spells were added; a per-card change log accompanies the data.

S-III.B Design regularities in full

Fig. S1 gives all three panels of Fig. 1; the main text shows (a) and (b).

Play Timing Is Free; Choice Is Priced: For 47 spells whose effects are fully priced, grouping by timing keyword and comparing cost minus priced effect value (a positive value means the attribute is charged for) gives Table S7. Cards with timing keywords pay nothing extra; the most flexible group (Reaction) has the *lowest* mean (-0.37 C), and the 95% interval for all keyword-bearing spells is -0.61 to $+0.36$ C. Same-effect contrasts show no price difference



(a) Number of triggers t

(b) Implied activations n^* (132 units, bins of 0.25)

(c) Implied realisation fraction f^* (46 conditional bonuses, bins of 0.25)

Fig. S1. The 50% rule and its limit. (a) Median binned value/cost ratio $\eta_{(t)}$ of the 162 units with a repeatable trigger, by number of triggers booked (dashed line: par, 1.00); the curve crosses 1.00 near half a trigger, and the AMBER score (Section III-D) adopts this 0.5-activation baseline as its ranking convention. (b) Implied activations $n^* = (C_{\text{tot}} - V_0)/(V_1 - V_0)$ for the 132 units whose single trigger is worth at least 1 C (bins of 0.25, left-closed; the dashed line marks 0.5). The distribution is spiky: 40 units sit exactly at 0 (9 of the 17 conditional triggers among them), 20 at 0.5 and 15 at 1. (c) Implied realisation fraction $f^* = (C_{\text{tot}} - V_B)/\Delta_{\text{cond}}$ for the 46 fully priced cards whose one-off conditional bonus is worth at least 1 C (bins of 0.25, left-closed; dashed line 0.5). Unlike n^*, the mass is at zero (20 cards exactly at 0) with a second group at 0.5–0.8, most of it cards whose condition the player controls; the base scenario books these bonuses at zero (Section III-D).

either: Vengeance vs. Blast of Power, Falling Star vs. Incinerate, Progress Day vs. Downstage Dramatics. By contrast, modal cards pay ≈0.8 C above their best mode (App. S-II.C, Modal).

This is not a null result attributable to noise. A wider play window is a strict advantage in game-theoretic terms, and in other CCGs instant-speed effects usually carry a body or effect discount. In Riftbound the advantage is uncharged. The plausible design rationale is that Reaction and Action are used as domain-identity tools and to encourage interactive, second-mover play, and are deliberately kept free of a cost gate. It is a defining feature of this game's cost model: numbers buy might and effects; timing is a free qualitative tag. The one exception is a this-turn might change ($\pm X$) or stun on a spell *without* any timing keyword, which shares the unit's inability to act inside a showdown and is booked at 0.5 (only *Onslaught* (VEN-081) calibrates it; low confidence). That factor enters Table S7 only through Onslaught, the one fully priced spell without a timing keyword that carries such an effect;

TABLE S6
CARD POOL, VALUATION COVERAGE AND SCOPE OF THE PRICED SAMPLE

(a) Coverage by card class

Class	Type	n	Full	Partial	Deferred
Costed; η defined, ranked, ablated	Unit	469	359	51	59
	Spell	233	138	19	76[a]
	Subtotal	702	497 (70.8%)	70 (10.0%)	135 (19.2%)
Costed; expected-uses model, not ablated	Gear[b]	107	69	19	19
Zero cost; no η, not ranked	Legend	49	46 per-use net values		3
	Battlefield	64	61 per-trigger yields[c]		3[d]
Excluded	Reminder card	1	Buff-token reminder, not a playable card		

(b) Fully priced, partially priced and deferred units and spells

	Full $n = 497$	Partial $n = 70$	Deferred $n = 135$	p
C_{tot}, median (IQR)	4 (3–6)	5 (4–7)	5 (3–8)	< 0.001
Spells, %	27.8	27.1	56.3	< 0.001
Set, %				0.008
Origins	31.4	24.3	39.3	
Spiritforged	20.9	12.9	28.9	
Unleashed	25.4	31.4	16.3	
Vendetta	19.5	28.6	12.6	
Proving Grounds	2.8	2.9	3.0	
Domain, %				0.001
Fury	17.9	12.9	9.6	
Calm	15.1	17.1	15.6	
Mind	15.9	8.6	14.1	
Body	15.3	12.9	17.0	
Chaos	13.9	24.3	17.0	
Order	17.5	17.1	9.6	
Two domains	4.4	7.1	14.1	
None (tokens)	0.0	0.0	3.0	
Rarity, %				< 0.001
Common	42.7	11.4	13.3	
Uncommon	22.5	22.9	21.5	
Rare	20.7	41.4	37.0	
Epic[e]	14.1	24.3	28.1	

[a]Deferred units: effects on all or an indefinite number of targets (16), gear- or legend-dependent effects (12), effects scaling with might (12), play from the trash (6), zero-cost tokens (4), rule-changing effects (4), free play of other cards (2), TEMPORARY or copy effects (2), deferred by decision (*Jinx, Demolitionist* (OGN-030)) (1). Deferred spells: effects on all or an indefinite number of targets (21), effects scaling with might (16), counters (11), symmetric effects (7), control theft (5), effects with no current price (3), banned cards set aside (3; Section VI-D), gear- or legend-dependent effects (3), play from the trash (2), free play of other cards (2), rule-changing effects (1), untargetability (1), TEMPORARY or copy effects (1). Each card is deferred by the first applicable rule; the per-card list accompanies the data.
[b]106 gear cards plus one unit gear (*Patched Porobot* (VEN-058)).
[c]14 direct, 19 conditional, 25 symmetric (both players), 3 controller cost reductions.
[d]2 banned (set aside; Section VI-D), 1 not computed.
(b): percentages are within column. p: Kruskal–Wallis for cost; for categories, χ^2 test of independence with a permutation p-value (20 000 draws; the asymptotic p-values agree to within 0.001). The four zero-cost tokens are excluded from the domain test. [e]Includes three alternate-art cards.

TABLE S7
EXTRA COST OF TIMING KEYWORDS OVER 47 FULLY PRICED SPELLS

Group	n	Mean (cost − value), C
No timing keyword	11	−0.09
ACTION	14	−0.14
ACTION + HIDDEN	5	−0.28
REACTION	12	−0.37
REACTION + REPEAT / HIDDEN	5	0.00

scoring it at full price instead would move the no-keyword mean by at most 0.3 C, within the interval above.

Diminishing Selection Premia: PREDICT, look-N-take-1 and modal effects are all expected-maximum problems, and their premia are concave in the number of options. Moving from two to three modes adds ≈0.4 C, to four ≈0.25 C, to five ≈0.2 C; for look-N-take-1 the increments from $N = 2$ to 5 are ≈0.28, 0.17 and 0.12 C. Within the common range of 1–5 cards, PREDICT is well approximated linearly at 0.40 C per card.

XP Is Priced at Both Ends but Hidden by Rounding: Fitting the gain-side price over all 18 XP-gaining units (might at $\lfloor B(c) \rfloor$; repeatable gains such as HUNT and level conditions discounted by the 50% rule) gives 0.38 C per point by through-origin least squares (0.59 C without rounding); we adopt 0.40 C, close to the spend-side 0.50 C. The sample is bimodal: 12 cheap or at-par units imply ≈0, and 6 units with might below cost (*Herald of Spring* (UNL-034), *Mister Root* (UNL-127), *Scorchclaw* (UNL-016), *Mosstomper* (UNL-047), *Gustwalker* (UNL-075), *Kha'Zix, Mutating Horror* (UNL-143)) imply ≈1 C per point. The former are not unpriced: on 2–5 C units the budget slack is under 1 C and 0.4 C of XP is absorbed by rounding. Level-6 and level-11 conditions show negative residuals, suggesting high thresholds are booked below 50%. Because XP has positive feedback, XP cards that reach their thresholds tend to exceed their cost, making them the natural place to look for outliers in future sets.

Cheap Units Get On-Play Effects Almost Free: Stratifying the 25 single-on-play-effect units by cost (Table S8) shows a sharp layer: 2–3 C units price the effect at roughly 30% of spell price regardless of type (best fit 0.26 with the combat-trick factor fixed at 0.5), while units at 4 C and above pay full price. Integer might is the mechanism—*integer-rounding absorption*: attaching a 1–4 C effect to a cheap unit removes 0 or 1 might, so most of the effect is a gift. When the same effect is sold as an optional additional cost on a cheap unit, it is priced at 0.5 or more, so the designer is not shrinking cheap bodies to pay for small effects. The factor is an average;

TABLE S8
RECOVERED ON-PLAY EFFECT FRACTION BY UNIT COST (25 UNITS)

	2–3 C	≥ 4 C
Combat-trick type	5 cards, mean 0.12 (Teemo, Scout; Gem Jammer; First Mate; Bubble Bot; Solari Shieldbearer)	2 cards: Field Musicians (VEN-026) 0.33, Whiteflame Protector 0.75
Other on-play	5 cards, mean 0.27 (Tideturner; Pit Rookie; Evershade Stalker; Lecturing Yordle; Windsinger)	13 cards, median 1.00

individual cards scatter (*Scuttle Crab* (UNL-053) has 2 fewer might than cost and its on-play draw is at nearly full price). The factor applies to effects included in the printed cost; it is not applied to optional-additional-cost effects, tokens, negative cost-type effects, or single-anchor cards such as Bewitching Spirit; keywords follow the separate rule below. Applying this rule returns a cluster of 2–3 C units previously at $\eta = 1.25$–2.00 to near 1.

Keyword rounding. Keywords that a 2–3 C unit carries natively are rounded in the same way, but only when their total price is below 1 C. Of the nine such units carrying VISION (0.40 C), ASSAULT 1 or SHIELD 1 (0.50 C), seven have might equal to cost, and the two that give up one might (*Black Rose Dignitary* (UNL-152) and *Lucian, Gunslinger* (SFD-028)) do so for a Deathknell and an attack trigger; at full keyword price the seven sit at $\eta = 1.17$–1.25, Black Rose Dignitary at 1.17 (with its Deathknell at $\lambda = 0.5$) and Lucian, whose attack trigger lies outside the base value, at 0.83. Keywords worth 1 C or more are paid for in might by five of the six 2–3 C units that carry them (e.g. *Inferna* (UNL-002), Assault 2; *Pyke, Dockside Butcher* (UNL-028), GANKING). The model therefore books sub-1 C native keywords on 2–3 C units at 0.30 of their price (VISION 0.12 C, ASSAULT 1 or SHIELD 1 0.15 C), which returns the seven units to 1.05–1.07 (Black Rose Dignitary 1.05, Lucian 0.72), and books keywords worth 1 C or more in full; keywords gained on a condition or on EMPOWER, or granted to other units, are not rounded. *Laurent Bladekeeper* (SFD-096) (3 C, 3 might, GANKING) is the one unit that does not pay for a 1 C keyword; at full price its $\eta = 1.33$ is recorded as a genuine excess and it is not used as an anchor. The rule stops at 3 C because of the budget slack $B(c) - c$: at 2–3 C it is 0.2–0.3 C, less than the keyword's price, so the keyword exceeds the budget and is a gift; from 4 C the slack (≥ 0.4 C) pays for it, and at full keyword price the 4–6 C units sit on or under the budget line (*Jhin, Meticulous Killer* (UNL-089) at $4.40 = B(4)$, *Petty Officer* (OGN-215) at $5.50 = B(5)$), with *Jeweled Colossus* (OGN-086) 0.4 C above it. Extending the rule to 4 C and above would discount twice and raises the full model's leave-one-out RMSE from 1.545 to 1.552 (on the v5 table, on which this check was run). The boundary is the one place where the unrounded budget slope matters (Section IV-C): with a slope of 1.05 the 4 C slack would be 0.2 C and would not cover a keyword, and with 1.15 the 3 C slack of 0.45 C would cover VISION. The two groups of units behave alike (neither gives up might for the keyword) and the out-of-sample difference between stopping at 3 C and at 4 C is below resolution, so the 3 C boundary is a convention that follows from the slope 1.1 rather than a finding.

Flow Is Booked on the First Play: The two plays of a Flow card have different cost structures; the saved hand card (1.00 C) is realised on the Flow play, so with $F = C_{\text{tot}}$ the Flow play is the more efficient. The premium the designer books on the printed side for the keyword is small (0.2 C adopted, range 0–0.6 C; App. S-II.C, Flow) and vanishes once $F \geq C_{\text{tot}} + 1$.

Scenario Effects at About Six Tenths: Unconditional gear removal is priced from four anchors: *Brittle Steel* (VEN-003) (2 energy + 1 power = 4 C; Flow cost 2 C above printed, so $k = 0$) 4.00; *Disarming Rake* (SFD-032) (5 C unit, 2 might, may kill a gear on play) $B(5) - 2 = 3.50$; *Salvage* (OGN-224) (4 C, kill a gear and draw 1) 2.00 after the draw; *Turn to Dust* (UNL-070) (2 C, gear becomes TEMPORARY) 2.00, a lower bound because the kill is delayed. Median 2.75 C (mean 2.88); we adopt 2.75. As a magnitude check only—not a derivation—kill-a-unit (8 C) is about $1.6\times$ the median unit total cost (5 C); the same multiple on the median gear cost (3 C) gives ≈4.8 C, and $2.75/4.8 \approx 0.57$ coincides with the free-Empower factor (median Empower cost 4 C $\times 0.6 = 2.4$ C; *Wild Claw* (VEN-089) recovers 2.36 C). A reasonable reading is that both are scenario-dependent effects—the opponent does not always have a gear or an Empower target—that the designer books below full value. The ≈0.6 factor is an empirical observation, not a consequence of the magnitude check.

Deathknell Booking by Set: A Hypothesis: To test whether λ varies by set, we add units whose Deathknell creates tokens (tokens at might $+0.40$) and *Vendetta* units that gain Deathknell on Empower (Empower cost at 1:1, other Empower benefits deducted). Table S9 gives per-set fits. The supplementary cards and their implied coefficients are *Machine Evangel* (OGN-239) $(8-4)/4.2 = 0.95$ and *Ferrous Forerunner* (SFD-021) $(9-6)/6.8 = 0.44$ (two 3-might tokens), *Carrion Dredger* (UNL-153) $(2-1)/1.4 = 0.71$, *Noxian Emissary* (VEN-128) $3/2.8 = 1.07$ (Empower 1 Order, 3 C, for two 1-might Recruit tokens) and *Baccai Witherclaw* (VEN-078) $3/4 = 0.75$ (Empower 5 C less the $+2$ might it also grants, for two runes). Point estimates trend upward with release order (0.53, 0.44, 0.69, 0.86; fitted line from ≈0.52 to ≈0.80), but a linear regression of λ on release order over all 10 samples has slope $+0.093$ per set (s.e. 0.060), Spearman $\rho = 0.44$, one-sided permutation $p \approx 0.10$; pooling *Origins+Spiritforged* (mean 0.55) against *Unleashed+Vendetta* (mean 0.74) gives a difference of 0.20 with 95% CI -0.14 to $+0.54$. Table S9 and the regression use Black Rose Dignitary at $\lambda = 0.43$ (Assault 1 at the keyword-rounded 0.15 C); at the full keyword price it would be 0.25, which slightly weakens the trend without changing the conclusion. None of the three tests rejects a common λ, so "later sets book Deathknell more conservatively" is recorded as a hypothesis, not a finding; the model uses $\lambda = 0.50$ for all sets. If confirmed, early cards such as Tasty Faefolk would be over-budget at the unit level and later cards such as Ruined Rex ($\eta_B = 0.84$) slightly under-priced by a uniform λ. Caveats: the late-set sample is five cards, the two

TABLE S9
DEATHKNELL COEFFICIENT λ BY SET (RELEASE ORDER)

Set	n	λ (LS / median)
1. Origins	4	0.53 / 0.50
2. Spiritforged	1	0.44
3. Unleashed	3	0.69 / 0.71
4. Vendetta	2	0.86 / 0.91
All (single-Deathknell units only)	5	0.46 / 0.50
All (with token and Empower samples)	10	0.58 / 0.61

Vendetta cards are recovered through the Empower route, and the excluded outlier Undercover Agent ($\lambda = 1.50$) would raise the *Origins* mean if included.

Ban Status Is Not Evidence of Single-Card Excess: Bans usually address combinations, and ban status is not a pricing criterion (Section IV-E). Twelve cards are banned in Standard constructed as of 24 September 2026 (nine since 31 March 2026, *Stealthy Pursuer* (OGN-177) since 24 July, *Ekko, Recurrent* (OGN-110) and *Stacked Deck* (OGN-183) since 18 September). The seven costed ones span the AMBER scale rather than its upper tail. Stacked Deck (1 C, look-3-take-1) scores 2.83—the same value as the unbanned *Lightning Rush* (VEN-156), which trashes rather than recycles the other two cards; *Called Shot* (SFD-122) 1.28; *Draven, Vanquisher* (SFD-020) 1.11 (1.00 nominal); *Scrapheap* (OGN-182) 1.00 (gear; its second draw, on discard or kill, is conditional); *Fight or Flight* (OGN-168) 1.00 (2 C to move a unit from a battlefield to its base, with HIDDEN and ACTION at zero); Stealthy Pursuer 0.75 (0.67 nominal: 6 C for 4 might and a move-along trigger); and Ekko, Recurrent is deferred because its Deathknell readies all runes (0.71 on the priced part, 5 might for 7 C). Of the five banned battlefields, *Aspirant's Climb* (OGN-276) changes a rule, *The Arena's Greatest* (OGN-290) and *Obelisk of Power* (OGN-284) are symmetric (a point, or a rune, to each player), *Reaver's Row* (OGN-285) yields one friendly move per defence and *The Dreaming Tree* (OGN-292) one draw per turn to whoever meets its condition. What the bans have in common—loops, deck-thinning consistency, tempo and first-player effects—lies in the value-fairness layer that the score does not measure. Three of the banned spells and two of the battlefields had been set aside under an earlier version of the deferral rules and remain outside the statistics so that the preregistered frozen model is unchanged (Table I); the other seven carry no such flag and are priced like any other card.

S-III.C Cross-predictions, ablation, paired trigger test and parameter review in full

Ablation: The mechanism models were introduced incrementally; Table S11 measures what each adds. We compare four nested models—A, linear additive core only ($C_{\text{tot}} = E + 2P$, Section IV-A; (3), (4)); B, A + mental accounting of hand cards (Section IV-D, closing draw/discard/Flow); C, B + order-statistics selection premia (Section IV-F); D, C + cost-controlled on-play adjustments (App. S-II.C, On-Play Effects)—and report RMSE, MAE and R^2 of V against C_{tot}, with residual $r_i = V_i - C_{\text{tot},i}$ on base value, over the 497 fully priced units and spells. D is the full model, and its values are those of the current value table; it also applies keyword rounding (Section VI-C), which is part of layer D. Removing a layer applies the counterfactual rule of the smaller model: A and B credit PREDICT at 0.60 C per card (Clairvoyance's 7 C less two draws, over five cards), price look-N-take-1 as a single draw and modal effects at their best mode without premium (worst mode when the opponent chooses); A books no Flow keyword premium; A–C price units' on-play effects and keywords at full spell price. Discard is -1.00 C, the non-showdown factor on spells is 0.50, and the same-target complementarity premium (Section V-C) is booked in every model, since none of these is an ablated mechanism. Every price whose calibration cards use a removed mechanism is re-estimated on the same cards with the same statistic; in practice only the token premium moves (0.40 to 0.50 C under A, because two of its thirteen calibration spells carry Flow).

Under leave-one-out cross-validation [31], each card is held out, every anchor-derived price is re-estimated from the remaining cards with the same statistic and chaining (a price adopted as a rounded statistic is shifted by the change in that statistic; the token premium and the Flow coefficient calibrate each other and are iterated to a fixed point), and the held-out card is then predicted. Sixteen cards cannot be predicted because a price they use is identified by that card alone—for example *Rune Prison* (OGN-050) for an unrestricted stun, Clairvoyance for PREDICT, Onslaught for the non-showdown factor—or, for *Guerilla Warfare* (OGN-264), by a card-specific value; they are dropped from all four models, leaving 481 cards (354 units, 127 spells). Two other card-specific values are replaced by the nearest global price when their card is held out: *Sacrifice* (UNL-173) takes the ally-sacrifice price (-2 C) and *Turn to Dust* (UNL-070) the median of the other three gear-destruction cards.

Only layer D reduces error materially. Over all cards it lowers the sum of squared residuals (SSE) by 49.7 relative to C (RMSE 1.537 to 1.504, R^2 0.665 to 0.679); of the 26 units it changes, 22 move closer to their cost, 3 move away and 1 is unchanged. The gain survives cross-validation almost intact (SSE -49.3; RMSE 1.574 to 1.541, R^2 0.650 to 0.665), with 25 of the 29 changed units improving, so the low-cost and combat-trick factors are not artefacts of their own calibration cards. D leaves spells unchanged by construction.

Layers B and C change few base values, and their effect is at the resolution limit. C alters 21 cards and lowers SSE by 1.2 in-sample (1.5 on the cross-validated subset, and the same out of sample), but its MAE is marginally higher than B's (0.9103 against 0.9099) and it fits spells slightly worse, mainly because the look-3-take-1 premium raises the residual of *Lightning Rush* (VEN-156) from $+1.00$ to $+1.83$ C. B alters 16 cards and lowers SSE by 0.5 in-sample; on the 481-card subset the change is -0.2 in-sample and -0.1 out of sample (spells $+0.1$ in both), i.e. within resolution in either direction. The Flow coefficient rests on four calibration spells, and its leave-one-out cost is the variance of that estimate rather than evidence against the keyword: with k fixed at 0.2 the B layer's out-of-sample SSE is lower than with $k \equiv 0$ (by 0.6 before the complementarity premium of Section V-C was booked), and no

TABLE S10
CROSS-PREDICTIONS BETWEEN SEPARATELY CALIBRATED PARAMETERS (ROWS MARKED C: CONSISTENCY CHECKS, NOT HOLD-OUTS)

Quantity predicted	Calibrated from	Prediction	Observed
(C) Discard the lowest-quality card in a hand (own -1.00; opponent's, their choice, $+1.00$)	Order-statistics model with the jointly fitted dispersion	0.83–1.04 C	1.00 C (Lunar Boon, Chemtech Enforcer; Bewitching Spirit for the opponent's hand)
Zero hand-rent intercept (playing is rent-free)	Damage pair Incinerate (2 C, deal 2) / Final Spark (8 C, deal 8): 1 C per point, intercept 0	Other effect lines pass through the origin	Draw: Consult the Past (4 C, draw 2), Progress Day (8 C, draw 4) give 2 C per card, intercept 0
(C) 1 power	Three spell equations	2 C	Channel 1 exhausted rune = 2 C (Mobilize, Catalyst of Aeons), reading a rune as a unit of power
One free Empower	Median Empower cost $\times 0.6$	2.4 C	Wild Claw recovers 2.36 C
Ready 2 C, buff 1 C, return-from-trash 2 C	Individual anchors (Upstage Comedy, Wallop; Morbid Return)	base $\eta = 1.00$	Call to Glory (OGN-207), Confront (OGN-129), Rally the Troops (SFD-166), Here to Help (SFD-111), Undying Loyalty (UNL-168) all 1.00
Opponent chooses the worse mode	Modal model (expected minimum)	3.84 C	Shakedown printed 4 C
Remove damage + recall + exhaust bundle	Zhonya's Hourglass (OGN-077), Tactical Retreat (UNL-175)	$\approx$2 C	Soraka, Wanderer: cost − might = 2 C for the same bundle as a trigger
May be played directly to a battlefield	Sneaky Deckhand, Rengar, Pouncing (anchors)	$+1$ C	Ocean Drake 1.00 C, Sai Scout 0.60 (units not used as anchors); Dragon Roost battlefield yield, valued as one friendly move, 1 C

estimation rule for k predicts held-out Flow cards better than any other (Section VII-C). Under cross-validation the spells are fitted about equally by all four models (RMSE 1.489; MAE 0.842–0.847, lowest under A). The small size of B and C is expected: discard is priced identically in every model, only five Flow cards have $p = 0$, and selection premia enter 21 base values.

Because 436 of the 497 cards take the same value under all four models, full-sample metrics dilute any layer that touches few cards. Table S12 therefore repeats the comparison on the cards each layer changes. On its own 26 cards (29 under leave-one-out), D lowers RMSE from 1.46 to 0.46 C in-sample and from 1.38 to 0.45 C out of sample; both 90% bootstrap intervals exclude zero, and 22 of 26 (25 of 29) cards move closer to their cost. B and C stay indistinguishable from the layer below even where they act. On B's 16 cards RMSE falls by 0.028 C in-sample and 0.003 C out of sample; on C's 21 (20) cards by 0.027 (0.035) C. Both intervals include zero in both evaluations, nearly as many cards get worse as better (9/7 and 8/7 for B; 12/9 and 12/8 for C), and the paired signed-rank tests give $p \geq 0.61$. The small full-sample effect of B and C is therefore not only dilution: on the cards they change, their corrections are of the order of the per-card noise. With 16–29 cards per subset these results are descriptive and are not read as significance tests of the mechanisms.

Printed costs move in 1 C steps, so a layer could be informative without lowering squared error if it moved cards onto their printed integer. We therefore also counted, on the same subsets, the cards whose value rounds to the printed cost ($|V - C_{\text{tot}}| < 0.5$) before and after each layer, with an exact McNemar test (one-sided, for an increase). The count is a valid check for D, the positive control: 5 of its 26 cards hit under C and 18 under D (15 gained, 2 lost; $p = 0.001$; 8 to 21 of 29 under leave-one-out, $p = 0.001$). For C the direction is the same but the difference is not significant: 5 of 21 cards hit under B and 8 under C (5 gained, 2 lost; $p = 0.23$; 4 to 8 of 20 out of sample, $p = 0.11$). For B the count is uninformative: 10 of its 16 cards sit exactly ± 0.50 C from their cost under A, B moves them by 0.1 C, and whether that counts as a gain depends entirely on the tie convention (7 gained under a strict inequality, none under an inclusive one). With 16–21 affected cards the test has little power, and it does not change the reading above: whether the designer uses the hand-card and selection rules cannot be decided at the current sample size and 1 C cost granularity; we retain both layers for interpretability and because their cross-checks hold (Table III).

The overall level of error comes from what no layer addresses. Base values exclude conditional and triggered effects, so residuals are negative on average (-0.57 to -0.62 C in-sample, -0.58 to -0.63 C out of sample) and in-sample R^2 stays between 0.66 and 0.68 in every model. Cross-validation raises RMSE by 0.012–0.013 C in every model, mostly through Sacrifice ($+3.00$ C when held out) and the four gear-destruction cards. The ablation, like the preregistered benchmark, uses base values; booking repeatable triggers as the AMBER score does removes part of this bias (Section VII-B).

Trigger Booking as a Paired Test: Table S13 tests the trigger factors of Section III-D on the full model D under leave-one-out and, in its last two rows, the booking of one-off conditional bonuses (Section VI-A). Each trigger version adds $k(V_1 - V_0)$ to the held-out prediction of the in-scope units: *nominal* ($k = 0$), *uniform* ($k = 0.5$ for all 162 units), *baseline* (uniform, with $k = 0.25$ for the 17 conditional triggers; the AMBER score), and two versions with a coefficient per trigger type (the leave-one-out median of n^* within the type), without and with the conditional halving. Each pair is compared on the squared errors of the cards whose k differs, by a paired bootstrap over cards (2000 resamples), a sign-flip permutation test of the mean difference (10,000) and the Wilcoxon signed-rank test (exact below $n = 25$), with seed 20260924. Leave-one-out re-estimates the effect prices in V but not the factors themselves (removing any one card leaves the median n^* at 0.5), so the table asks whether the fixed factors improve prediction card by card; it is not an out-of-sample test of the factors. That test is H9 of Section VIII.

Both steps of the baseline convention improve prediction card by card. Uniform booking brings 95 of the 162 units closer to their cost (66 further) and lowers the leave-one-out RMSE over all 481 cards from 1.541 to 1.438 C; the conditional halving brings 14 of its 17 cards closer and lowers

TABLE S11
ABLATION: IN-SAMPLE AND LEAVE-ONE-OUT ERROR OF V AGAINST C_{tot}

	In-sample			Leave-one-out		
Model	RMSE (C)	MAE (C)	R^2	RMSE (C)	MAE (C)	R^2
All fully priced cards (in-sample $n = 497$; LOO $n = 481$)						
A: linear core	1.538	0.910	0.664	1.575	0.957	0.650
B: A + hand-card mental accounting	1.538	0.910	0.665	1.575	0.957	0.650
C: B + order-statistics premia	1.537	0.910	0.665	1.574	0.957	0.650
D: C + on-play cost adjustments (full)	1.504	0.871	0.679	1.541	0.916	0.665
Units (in-sample $n = 359$; LOO $n = 354$)						
A: linear core	1.592	0.980	0.663	1.605	0.998	0.656
B: A + hand-card mental accounting	1.592	0.979	0.663	1.605	0.998	0.656
C: B + order-statistics premia	1.591	0.977	0.663	1.603	0.996	0.657
D: C + on-play cost adjustments (full)	1.547	0.923	0.682	1.559	0.941	0.676
Spells (in-sample $n = 138$; LOO $n = 127$)						
A: linear core	1.388	0.728	0.556	1.489	0.842	0.506
B: A + hand-card mental accounting	1.387	0.730	0.556	1.489	0.844	0.506
C: B + order-statistics premia	1.388	0.736	0.556	1.489	0.847	0.506
D: C + on-play cost adjustments (full)	1.388	0.736	0.556	1.489	0.847	0.506

Residual $r_i = V_i - C_{\mathrm{tot},i}$ on nominal (base) value; $R^2 = 1 - \sum r_i^2 / \sum (C_{\mathrm{tot},i} - \bar{C}_{\mathrm{tot}})^2$. Leave-one-out: each card is held out and every anchor-derived price is re-estimated without it under the same layer (including chained re-estimates), then the card is predicted. Sixteen cards whose price is identified only by the card itself (single-anchor prices and one card-specific value) cannot be predicted and are dropped from all four layers; in-sample error on the same 481 cards is RMSE 1.563 / 1.563 / 1.562 / 1.528 and MAE 0.938 / 0.938 / 0.938 / 0.897 for A / B / C / D. Values are those of value table v6 (the same-target complementarity premium of Section V-C is booked in every layer). Known effect of D over C on units (nominal ratio): $\eta \geq 1.5$ from 8 to 1, $\eta > 1.17$ from 29 to 13 (with keyword rounding); median and quartiles unchanged (1.00; 0.75–1.00).

TABLE S12
ABLATION ON THE CARDS EACH LAYER CHANGES

Layer	Eval.	n	RMSE (C)	ΔRMSE [90% CI]	$+/-/=$	p
A→B	In	16	0.631→0.604	−0.028 [−0.07, +0.03]	9/7/0	0.81
	LOO	16	0.598→0.594	−0.003 [−0.03, +0.03]	8/7/1	0.95
B→C	In	21	1.057→1.030	−0.027 [−0.15, +0.11]	12/9/0	0.82
	LOO	20	1.083→1.048	−0.035 [−0.16, +0.10]	12/8/0	0.61
C→D	In	26	1.456→0.455	−1.001 [−1.39, −0.59]	22/3/1	<0.001
	LOO	29	1.380→0.453	−0.927 [−1.30, −0.53]	25/3/1	<0.001

Each subset holds the cards whose value differs between the two layers under that evaluation (In: in-sample, 497 cards; LOO: leave-one-out, among the 481 predictable cards). RMSE is of $r_i = V_i - C_{\mathrm{tot},i}$ under the lower and the upper layer. $+/-/=$: cards whose $|r_i|$ decreases, increases or is unchanged. CI: percentile bootstrap over cards (10,000 resamples). p: two-sided exact Wilcoxon signed-rank test on paired $|r_i|$ (zero differences dropped, mid-ranks for ties).

it to 1.400 C (in-sample 1.504 to 1.365 C, mean residual -0.62 to -0.29 C). Both intervals exclude zero on the changed cards and on all 481 cards, and the three tests agree. The conditional gain is concentrated: four high-increment cards (Swain, Yasuo, *Shen, Leader of the Kinkou Order* (VEN-138) and *Tryndamere, Barbarian* (OGN-034)) account for 90% of its reduction in squared error, and on the 17 cards a factor of 0.25 predicts better than 0 (RMSE 0.818 against 1.471 C). Per-type coefficients are not adopted (Section VII-C): against the uniform and baseline versions their intervals include zero, and on the 128 unconditional cards they change, they predict worse than the uniform factor.

The same test on one-off conditional bonuses (last two rows) finds no factor to adopt. Booking Δ_{cond} at 0.5 brings 23 of the 53 cards closer to their cost and 30 further, with intervals that include zero on the changed cards and on all 481; booking it in full is significantly worse than 0.5 (8 closer, 43 further). The grid minimum lies at 0.25 (ΔRMSE -0.212 C, interval -0.391 to $+0.008$; p_{perm} 0.08, p_{W} 0.06), but four high-Δ_{cond} cards (*Solari Chief* (OGN-225), *Siphoning Strike* (VEN-146), *Master Yi, Meditative* (OGS-004), Poppy, Paragon) account for 72% of its reduction in squared error while 29 of the 53 cards move away from their cost, and the three tests disagree; it is not adopted. By set, a 0.5 booking helps on *Origins* and *Spiritforged* cards and hurts on *Unleashed* and *Vendetta* (median f^* 0.50, 0.25, 0.00 and 0.00): the opposite direction to the upward drift of λ in Section VI-D, and, like it, a per-set description rather than a rule. The AMBER score keeps $k_{\mathrm{cond}} = 0$.

Parameter Review: After the ablation, every price in the table was re-examined against the cards that use it. For each of the 37 multi-anchor or layer parameters we grouped the fully priced cards whose formula contains it, tested whether the clean cards among them (base value equal to full-trigger value, 199 cards) show a one-sided residual in-sample and under leave-one-out, and recorded how far the parameter moves when any one of its anchors is held out; parameters flagged by these screens were then examined within their own anchor set, comparing the leave-one-out error of alternative estimation rules and computing exact bootstrap intervals. The 15 prices identified by a single card were only cross-checked against a second card, never adjusted from residuals. A change was to be adopted only if the new value followed from the parameter's own anchors and comparable cards under the same statistic, differed from the current value by more than the 0.3 C resolution, and did not predict held-out anchors worse than the current value (or its interval excluded the current value).

No parameter met these conditions. The Flow coefficient is

TABLE S13
TRIGGER AND CONDITIONAL-BONUS BOOKING: PAIRED LEAVE-ONE-OUT COMPARISON UNDER MODEL D

Comparison	n	$+/-/=$	Changed cards RMSE (C)	Changed cards ΔRMSE [95%]	All 481 cards ΔRMSE [95%]	p_{perm} / p_{W}
Uniform 0.5 vs. nominal	162	95/66/1	1.793→1.518	-0.275 [-0.435, -0.102]	-0.103 [-0.171, -0.036]	0.002 / <0.001
Baseline vs. uniform	17	14/3/0	1.926→0.818	-1.108 [-1.751, -0.368]	-0.038 [-0.080, -0.008]	0.023 / 0.013
By type vs. uniform	139	71/66/2	1.395→1.361	-0.035 [-0.206, $+0.132$]	-0.010 [-0.060, $+0.037$]	0.70 / 0.94
By type + conditional vs. baseline	139	68/69/2	1.252→1.321	$+0.069$ [-0.062, $+0.207$]	$+0.018$ [-0.018, $+0.057$]	0.36 / 0.46
Conditional bonus at 0.5 vs. baseline (0)	53	23/30/0	1.316→1.209	-0.107 [-0.513, $+0.336$]	-0.011 [-0.055, $+0.037$]	0.67 / 0.61
Conditional bonus at 1.0 vs. 0.5	53	8/43/2	1.209→2.053	$+0.845$ [$+0.534$, $+1.128$]	$+0.105$ [$+0.049$, $+0.180$]	<0.001 / <0.001

Squared error of $V_{\text{LOO}} + k(V_1 - V_0)$ against C_{tot}. n: cards whose k differs between the two versions; $+/-/=$: cards closer to, further from, or as far from their cost under the first version. Leave-one-out RMSE over all 481 predictable cards: nominal 1.541, uniform 1.438, baseline 1.400, by type 1.429, by type + conditional 1.418 C. Intervals: paired percentile bootstrap, 2000 resamples. p_{perm}: sign-flip permutation, 10,000; p_{W}: Wilcoxon signed-rank, exact below $n = 25$ (baseline vs. uniform), tie-corrected normal approximation otherwise. Last two rows: $V_{\text{LOO}} + k(V_1 - V_0) + k_{\text{cond}}\Delta_{\text{cond}}$ on the 53 predictable cards with a one-off conditional bonus (Section VI-A), triggers at the baseline factors; seed 20260925. Leave-one-out RMSE over all 481 cards at $k_{\text{cond}} = 0$, 0.25, 0.5, 0.75 and 1: 1.400, 1.380, 1.390, 1.429, 1.495 C.

the case discussed above. Two screens that at first suggested changes were resolved by the might budget: the five 4–6 C units carrying a keyword worth less than 1 C (Section VI-C) give up no might for it, as the 2–3 C units do, but their keyword fits inside the budget slack $B(c) - c$, so extending keyword rounding to them would double-discount and worsens out-of-sample error (leave-one-out RMSE 1.545 to 1.552 on the v5 table, on which the check was run; Section VI-C notes that the boundary itself rests on the budget slope); and the $1.10\times$ multiplier recovered for on-play effects is the fractional slack under whole-card rounding (App. S-II.C, On-Play Effects). The prices of moving a friendly unit (1.00 C) and readying a unit (2.00 C) are each contradicted by a cluster of spells at $+1$ C, but the cluster is the same-target complementarity of Section V-C, not a mis-set price: any single value leaves four cards off by ± 1 C, while raising the ready price to 3 C would lower the full-sample cross-validated SSE by 10 only by lifting a set of expensive units that sit below the curve for unrelated reasons, at the cost of a 1 C error on both single-effect anchors. The gear-destruction price (2.75 C) rests on four anchors with a bimodal spread (2.0, 2.0, 3.5, 4.0; exact bootstrap 90% interval of the median 2.0–4.0), and the mean (2.875 C) is more stable under leave-one-out, but the difference is below resolution. Among single-anchor prices, the second card agrees for recurring a unit (*Cemetery Attendant* (OGN-165), *Undying Loyalty* (UNL-168)), preventing the next damage (*Tactical Retreat* (UNL-175)) and playing to an enemy battlefield (*Rengar, Trophy Hunter* (VEN-179)); it disagrees, without a way to decide, for moving an enemy to its base (*Maddened Marauder* (OGN-191) implies 1 C against Isolate's 2 C), an unrestricted stun (*Back Off* (UNL-042) implies at most 3 C, *Zenith Blade* (OGN-262) at least 6 C) and sacrificing an ally (*Sacrifice* (UNL-173) implies -5 C against Cruel Patron's -2 C; since resolved: Sacrifice alone asks for an ally of 5 might or more, and *Commander Ledros* (OGN-231), which takes 2 C off its cost for each friendly unit killed, confirms -2 C); these are listed in Table S5. Two methodological points follow. Parameters must not be tuned on the pooled residual of all cards that use them, because that residual is dominated by unpriced conditional effects and by cards that are simply weak; and unit residuals must be read against the right target—$\lfloor B(c) \rfloor$ for might, $B(c)$ for effects—

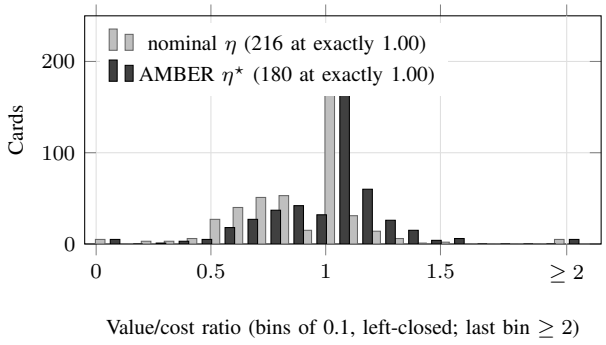


Fig. S2. Distribution of the nominal ratio and the AMBER score over the 497 fully priced units and spells. The bin $[1.0, 1.1)$ holds the cards at exactly 1.00, 146 of which are anchors; booking repeatable triggers at half an activation moves 55 units out of that bin to the right and 19 into it, and shifts the left tail of the 162 trigger units towards par.

since the difference between the two is exactly the size of a sub-1 C effect. The same review procedure, applied to a per-trigger-type coefficient for repeatable triggers (the leave-one-out median of n^* within each trigger type), found no stable gain: it does not predict held-out cards better than the uniform factor (ΔRMSE -0.010 C on 481 cards, 95% interval -0.060 to $+0.037$), and the low medians of move and conquer/hold triggers (0.25 and 0.27) rise to 0.43 and 0.45 once the 17 conditional triggers are removed (Kruskal–Wallis p from 0.027 to 0.13), so the apparent type differences are mostly the conditional effect (Section VII-B).

S-III.D Distribution of the score; highest-ratio cards and outliers

Table S14 lists the 15 highest AMBER scores among fully priced units and spells, with the nominal ratio alongside (ranks 4–5 tie at 2.00, 7–11 at 1.50 and 12–13 at 1.47). Eight are spells, seven of them this-turn might changes, and their scores equal their nominal ratios. The seven units enter through their repeatable triggers; ranked by the nominal ratio, the list instead ended with four spells and *Laurent Bladekeeper* (SFD-096) at 1.33 and *Tideturner* (OGN-199) at 1.30. Two of the four units at 1.50 or above, *Akali, Deadly Weapon* (VEN-021) and

Ambessa, Respected and Feared (VEN-136), have trigger tiers that presuppose they are EMPOWERED; their score uses the printed cost alone as denominator, a convention still under review. The two *Decree* spells owe their ratio to the 0.5 trait-restriction factor and are live only against one domain. Moonlight Affliction (-10, capped at 8) is 1.14; Onslaught (no timing keyword, $+6$ at the non-showdown discount) 0.80; *Back to Back* (OGN-206) (second target as conditional value) 0.67.

Several individually analysed cards illustrate the model. *Irresistible Faefolk* (UNL-112) has $\eta_{(1)} = 1.50$ and $\eta_{(3)} = 3.50$, against medians of 1.20 and 1.85 over the 162 trigger units (Fig. 1(a)); its nominal ratio agrees with its reputation among players. Tasty Faefolk (7 C, 6 might; 7 C should carry 8 might) trades 2 might for a Deathknell worth 6 C (two runes + draw 1), booked at 3 C: $\eta = 9/7 = 1.29$, $\eta_B = 1.10$, $\lambda = 0.33$. *Astral Heron* (VEN-044) (7 C, 7 might; cost reduction of 2 energy and 2 power on the next card each turn while at a battlefield, up to 6 C per trigger) has an implied effect price of only 1 C, 0.17 of the single-trigger value and far below the 3 C the 50% rule would imply; its $\eta_{(t)}$ is 1.00, 1.86, 2.71, 3.57 at $t = 0$–3. Under the baseline convention of Section III-D, Astral Heron scores 1.43, Irresistible Faefolk 1.00 (nominal 0.50), Stellacorn Herder 1.00 (nominal 0.75) and Traveling Merchant 1.25. Weakly priced cards include *Overt Operation* (OGN-153) ($\eta = 0.44$ with two buffed allies), *Rocket Barrage* (SFD-077) (0.72, about 1.5 C above the modal model's premium), *Shadow Dash* (VEN-148) and *Twilight Step* (VEN-105) (0.50 and 0.38; restricted moves, Flow costs 5 C and 2 C above printed), Brittle Steel (0.69; its 4.00 C gear-kill anchor exceeds the adopted 2.75) and Undercover Agent (0.86; $\lambda = 1.50$, excluded from the Deathknell fit). Perfect Execution, listed as weak in the first version of the table ($\eta = 0.74$), is at 0.94 once its ready-plus-Assault combination is priced (Section V-C). On the other side, the parameter review singles out *Wages of Pain* (SFD-070) (3 C for 3 damage plus a gold token; the token is effectively free), Lightning Rush (1 C for look-3-take-1, residual $+1.83$ C), *Decree of Focus* (VEN-040) and *Decree of Insight* (VEN-061) (1 C Decrees whose trait restriction is effectively worth 0.2–0.25 rather than 0.5, within the 1 C minimum-cost rounding) and *Jeweled Colossus* (OGN-086) (5 might with PREDICT and SHIELD 1 at 5 C, 0.4 C over the budget slack), and at 8 C *Zephyr Sage* (OGS-005), *Vanguard Attendant* (OGS-016) and *Garen, Rugged* (OGS-007), each two might below $\lfloor B(8) \rfloor$.

TABLE S14
TOP 15 AMBER SCORES AMONG FULLY PRICED UNITS AND SPELLS

#	Card	Type, cost	Nom. (η)	AMBER ($\eta^\star$)	Composition
1	Stupefy (OGN-095)	Spell 1 C	3.00	3.00	Reaction: -1 might (1) + draw 1 (2)
2	Lightning Rush (VEN-156)	Spell 1 C	2.83	2.83	Look 3, take 1 (2.83); rest to trash
3	Decree of Insight (VEN-061)	Spell 1 C	2.50	2.50	-5 might to a Body enemy (5×0.5)
4	Discipline (OGN-058)	Spell 2 C	2.00	2.00	Reaction: $+2$ might (2) + draw 1 (2)
4	Decree of Focus (VEN-040)	Spell 1 C	2.00	2.00	$+4$ might vs. a Fury enemy (4×0.5)
6	Trevor Snoozebottom (UNL-048)	Unit 3 C, 3 might	1.05	1.51	Base 3.15 $+ 0.5\times$ hold: play a 3-might Sprite (2.75)
7	Akali, Deadly Weapon (VEN-021)†	Unit 3 C, 3 might	1.00	1.50	3 might $+ 0.5\times$ move: damage (3)
7	Ambessa, Respected and Feared (VEN-136)†	Unit 5 C, 5 might	1.00	1.50	5 might $+ 0.5\times$ attack: kill a weaker enemy (5)
7	Nami, Headstrong (UNL-052)	Unit 3 C, 3 might	1.00	1.50	3 might $+ 0.5\times$ hold: ready and buff the next unit (3)
7	Cleave (OGN-004)	Spell 1 C	1.50	1.50	Action: Assault 3 this turn (1.5)
7	Grim Resolve (UNL-095)	Spell 2 C	1.50	1.50	Action: $+3$ might (3); XP on win is conditional
12	Eclipse (UNL-063)	Spell 3 C	1.47	1.47	Reaction: -4 might (4) + Predict (0.4)
12	Draven, Audacious (SFD-148)	Unit 8 C, 6 might	0.75	1.47	6 might $+ 0.5\times$ first combat win each turn: 1 point (11.5)
14	Ezreal, Prodigy (SFD-149)	Unit 5 C, 3 might	1.20	1.44	Base 6.00 (discard 1, draw 2) $+ 0.5\times$ cheaper optional cost (2.4)
15	Astral Heron (VEN-044)	Unit 7 C, 7 might	1.00	1.43	7 might $+ 0.5\times$ cost reduction on the next card (6)

Nom.: nominal ratio (triggers at zero); AMBER: triggers at 0.5 activations (0.25 under an independent condition; none of these 15). †Trigger tiers presuppose the unit is EMPOWERED; denominator is the printed cost only (under review).

APPENDIX S-IV. PREREGISTERED TEST: SENSITIVITY TIERS IN FULL

The main text reports three tiers of the blind test (Table V): the registered primary analysis, the registered exclusion of cards revealed before 25 September, and one post-hoc tier in which the five low-cost spells responsible for most of the set's spread are re-read under the revised rules of Section IX. Table S15 lists every tier that was computed after unblinding, each by the frozen scoring scripts called unchanged, together with the dispersion comparison of the set against the leave-one-out distribution of the earlier sets that motivated the re-reading. The preregistered numbers (tier 0) are reproduced exactly by the rerun. None of the tiers is part of the registration, and none enters H1–H10. Coverage of the set, as registered: of the 122 units and spells, 66 are fully priced (46 units, 20 spells), 32 partial and 24 deferred, and 60 of the 66 carry no low-confidence price; gear (4 of 27 rows fully priced, counting the BOMB token), legends and battlefields are reported descriptively only.

APPENDIX S-V. NEW KEYWORDS, THE STUN PRICE AND THE FIVE EXTREME SPELLS

S-V.A New keywords, the stun price and "can't move" in full

New keywords. DISARM N ("when I attack, give an enemy unit here $-N$ might this turn") is priced at 0.25 C per point: a joint least-squares fit on the four cards where it is the one unknown gives 0.26 (0.21 with a fifth, consistency-only card), and every leave-one-out refit lies between 0.08 and 0.39; the keyword-rounding rule of Section VI-C applies to it as to ASSAULT. SHOW OFF (reveal a card of a given kind as you play

TABLE S15
SENSITIVITY TIERS OF THE BLIND TEST AFTER UNBLINDING: H1 AND H2 RESCORED BY THE FROZEN SCRIPTS, AND THE DISPERSION OF THE SET AGAINST THE LEAVE-ONE-OUT EARLIER SETS (TIER 0 REPRODUCES THE REGISTERED RESULT; ALL OTHER TIERS ARE POST HOC)

Tier	n	H1 RMSE (90%)	H2 median	n_5 / n_{ctrl}	Dispersion ratio [95%]	p	IQR, set / LOO	Outside band, set / LOO	Nominal ratio [95%]
0: registered primary analysis (revised file, as published)	60	1.734 (1.372–2.065)	1.000	60 / 467	1.592 [1.106, 2.177]	0.009	0.382 / 0.250	51.7% / 40.3%	1.548 [1.035, 2.125]
1: the pre-check reading (two cards removed, Fanfare's triggers at one half)	58	1.699 (1.327–2.041)	1.000	58 / 467	1.354 [0.902, 1.951]	0.087	0.340 / 0.250	48.3% / 40.3%	1.410 [0.912, 1.988]
1b: as 1, with "can't move this turn" at -2 C	58	1.683 (1.307–2.026)	1.000	58 / 467	1.256 [0.870, 1.768]	0.198	0.340 / 0.250	48.3% / 40.3%	1.314 [0.881, 1.803]
2: strict v6.1 (Hunker Down back to partial)	59	1.729 (1.368–2.060)	1.000	59 / 467	1.427 [1.014, 1.928]	0.036	0.363 / 0.250	50.8% / 40.3%	1.386 [0.947, 1.864]
3: the five spells removed	55	1.653 (1.246–2.024)	1.000	55 / 467	1.016 [0.777, 1.275]	0.940	0.333 / 0.250	47.3% / 40.3%	0.982 [0.721, 1.241]
4†: v7 rules applied post hoc (reported in the main text); control group under the same rules	58	1.683 (1.307–2.026)	1.000	58 / 465	1.282 [0.890, 1.780]	0.166	0.340 / 0.243	48.3% / 40.0%	1.341 [0.914, 1.859]
4a: as 4, control group unchanged	58	1.683 (1.307–2.026)	1.000	58 / 467	1.256 [0.879, 1.780]	0.202	0.340 / 0.250	48.3% / 40.3%	1.314 [0.894, 1.813]

H1, H2: the frozen scoring script rerun on the tier's pricing file; all tiers leave H1 (≤ 1.841 C) and H2 (0.90–1.10) supported. Dispersion: Brown–Forsythe statistic, mean $|\eta^\star - \text{median}|$ of the set's primary cards over that of the leave-one-out earlier sets (467 cards; 465 in tier 4, where two count-dependent spells without a cap leave the control group and one is re-booked under the same rule), ratio with a stratified bootstrap interval (4000 resamples) and a type-stratified permutation p (10,000); IQR and the share of cards outside the empirical 0.86–1.17 band for the set and for the earlier sets; last column, the same ratio on the nominal score. Tier definitions: App. S-V. The registered secondary sample of all 66 fully priced cards gives H1 1.707 C (original file: $n = 62$, 1.742), both supported.

this) adds nothing by itself; the bonus it unlocks behaves exactly like the one-off conditional bonuses of Section VI-A, booked at zero on four of five cards and at 0.79 on *Rousing Display* (RAD-127), beside its near-twin *Vanguard Captain* (OGN-218) at 0.71. A BOMB token (a gear that deals 2 damage when destroyed) is worth 1 C, its Deathknell damage at $\lambda = 0.5$; the unit cards admit 0.7–1, while one spell and one gear point higher, so confidence is low. DEPLOY costs nothing: the two cards with only one-off effects are priced exactly at those effects, and the repeatable "here" effects of deploy gear are charged for about two uses, the median for ordinary conditional gear (Mann–Whitney $p = 0.28$); booking DEPLOY at -1 C is worse card by card (paired 95% interval of ΔRMSE -0.89 to -0.01). Gear that becomes a unit (*The Ball* (RAD-146)) has no reference card and is not priced.

Stun has fallen in price. The frozen model priced an unrestricted stun at 4 C from *Rune Prison* (OGN-050) and listed it as low-confidence, because *Back Off* (UNL-042) implied less and *Zenith Blade* (OGN-262) more. Read by set, the spells tell a consistent story: in the first set an unrestricted stun cost 4 C, a stun restricted to a unit in combat cost 2 C (*Facebreaker* (OGN-220), which also stuns an ally at zero), and from the second set on every spell on which stun is the one unknown is priced at 2 C (*Thwonk!* (SFD-040) and *Heroic Charge* (UNL-155), restricted; Ki Burst and One-Two Punch, unrestricted); *Existential Dread* (UNL-134), restricted, sits 1 C below its cost under any reading. Back Off (3 C) fits once its draw "if played from hand", a condition the player controls, is booked at one half. Signature cards are not used as anchors: Zenith Blade, Leona's signature spell, sits 2 C below its cost at the frozen stun price and 4 C below at 2 C, and the 20 fully priced signature spells, though not mispriced on average (mean residual -0.31 against -0.35 for other spells, permutation $p = 0.90$), range from -3 to $+2$ C, each priced for its champion; the signature unit *Daisy!* (UNL-196) is likewise set aside. On the nine remaining cards whose value changes, the staged price lowers the squared error from 12.2 to 7.3 (RMSE 1.16 to 0.90 C; paired 90% interval of ΔRMSE -0.77 to $+0.29$; three better, four worse, two unchanged), and the gain comes from Ki Burst and One-Two Punch. Over all eleven affected cards, signature cards included, the staged price is worse than the frozen one (ΔRMSE $+0.29$, -0.40 to $+0.85$), and the AMBER RMSE of the old pool rises from 1.365 to 1.377 C. The four that get worse are units whose stun comes from an attack or activated ability; they fit 2 C only if the ability is booked about once, so on them the stun price and the activation count cannot be separated.

Can't move is not exhaust. The two cards revised for "can't move this turn" fit better at -2 C than at the -1 C of an exhausted unit: *Entrenched Boar* (RAD-026), which may enter ready on that condition, is then priced exactly (ready and unable to move nets to zero), and *Hunker Down* (RAD-034) moves from $+2$ to $+1$ C, a remainder consistent with the surplus of other 1 C cards; the primary H1 error is 1.719 rather than 1.734 C. Exhausting a unit also costs its move, but in play the two differ, and the next version books "can't move this turn" at -2 C.

Other observations. Repeatable triggers stay at half an activation: on the 162 old-pool units a two-thirds booking is clearly worse (ΔRMSE $+0.14$ C, 90% interval $+0.05$ to $+0.25$), and on the fifth set the two cannot be separated. The median of the set's spells is slightly lower than in earlier sets (0.97, permutation $p = 0.07$ for the median difference, one of many comparisons and uncorrected), while their mean is higher (1.13, $p = 0.20$). These items, with the staged stun price and the conventions above, form the draft of the next model version.

S-V.B The five extreme spells of the blind test

The dispersion comparison of App. S-IV traced the wider spread of the fifth set to five low-cost spells, selected after unblinding by $|\eta^\star - 1|$. Each was re-derived from the card text and the frozen rules v6.1, with the blind pricing's formula

and the pre-unblinding revision list beside it, and then checked independently by a second AI session that received only the card text, the frozen rules, the frozen value table's precedents and the candidate readings. Table S16 summarises the outcome; the post-hoc tier reported in the main text (tier 4 of Table S15) applies the v7 readings of the last column. None of the five was a misreading of the card text; three expose gaps in the frozen rules that the blind test was designed to surface, one is simply strong, and one is champion-bound (a signature spell, which the next version does not price on its own). The preregistered numbers are left untouched.

Appendix S-VI. Baselines in Full

Table S17 gives the full comparison behind Table VI of the main text; Table S18 lists the card-face features of the ridge-regression baseline B2; the language-model baseline B3 is described at the end of this section.

The language-model baseline B3. A Claude Opus 5.5 sub-agent, started in a fresh workspace holding only a task file and the 60-card blind table (the blind pricing's card-face fields, without rarity), was asked to guess every card's cost to one decimal, with neither the rules, nor the price book, nor any earlier card's cost. As in the blind pricing, isolation rests on the instruction; the sub-agents' tool-call logs, showing reads of the task file and the blind table only, are corroboration. The first run was fixed in advance as primary; five further runs in separate workspaces followed, of which run 3 omitted RAD-078 (scored on 59 cards) and run 6 was added to keep five complete runs; the prompt, its hash and the raw outputs accompany the data.

Appendix S-VII. Per-Card Evidence for the Reverse-Engineered Conventions

Tables S19–S26 give, card by card, the evidence summarised in Table VII of the main text. Throughout, C is the printed cost C_{tot} in C; F is the value of the rest of the card under v6.1 (might, keywords and all other priced effects, the convention under study excluded); all per-card values are in C. For count-dependent effects, $N^* = (C - F)/(\text{value per instance})$ is the number of instances the printed cost pays for; where a unit of 7 C or more is read against the might budget $B(c)$ of Section IV-C, $N^* = (B(C) - F)/(\text{value per instance})$ for a bonus and $(C - B^{-1}(F))/(\text{value per instance})$ for a cost reduction. Leave-one-out estimates use, for each card, the median over the other cards of the same class.

TABLE S16
THE FIVE LOW-COST SPELLS BEHIND THE FIFTH SET'S WIDER SPREAD: BLIND VALUE, FINDING ON RE-READING, AND TREATMENT IN THE POST-HOC TIER

Card	C_{tot}	Blind V ($\eta^\star$)	Finding	Under v6.1 as frozen	Post-hoc tier (v7 reading)
Life from Stone (RAD-029)	1	3 (3.00)	Not an error: draw 1 (2) + channel a rune (2) − exhaust a friendly unit as an additional cost (1); the same formula as the 3 C earlier-set card of this shape	$V = 3$, fully priced	Unchanged (simply strong)
Hunker Down (RAD-034)	1	3 (3.00)	Rule gap: "can't move this turn" has no price in v6.1; the original blind file left the card partial, and the pre-unblinding revision booked the clause at -1 C by analogy with exhausting a unit (disclosed as deviation 9)	$V = 4$ on the priced part, partial (strict reading)	Clause at -2 C: $V = 2$
Fanfare (RAD-047)	3	0 (0.00)	Not an error: three "when you play a unit" triggers; v6.1 keeps spells with trigger bins at their nominal value ($k = 0.5$ applies to units only), as for the two earlier-set precedents	$V_0 = 0$, AMBER = nominal = 0	$k = 0.5$ extended to capped spell triggers: AMBER value 3 (V_0 still 0)
High Note (RAD-048)	3	0 (0.00)	Not an error: $-X$ might with X the number of exhausted units you control; one target, so the "indefinite number of targets" rule does not apply, and three earlier-set precedents are booked the same way at zero	$V = 0$, fully priced	Uncapped count: deferred (out of scope); the two uncapped precedents leave the control group, the capped one is re-booked at 2
Last Caress (RAD-154)	2	0 (0.00)	Not an error under v6.1: the whole effect waits for the next unit played this turn, a conditional booked at zero; the card is a signature spell (Evelynn), which v6.1 has no clause for	$V = 0$, fully priced	Signature spells not priced on their own: out of scope

TABLE S17
BASELINES IN FULL (FULL VERSION OF TABLE VI): (A) THE 60 PRIMARY CARDS OF THE FIFTH SET; (B) LEAVE-ONE-OUT ON THE EARLIER SETS

(a) Fifth set, 60 primary cards (ΔRMSE = ZHIZHI − baseline; negative favours ZHIZHI)

Method	RMSE	90%	MAE	Mean res.	Units (44)	Spells (16)	ΔRMSE	90%	p_{W} (n, z)	$+/-/=$
B0 constant	2.805	2.399–3.211	2.335	$+0.421$	2.830	2.736	-1.071	$[-1.450, -0.709]$	6.6×10^{-6} (60, -4.51)	48 / 12 / 0
B1 type × might	2.428	2.025–2.806	1.774	-0.815	2.306	2.736	-0.694	$[-1.043, -0.388]$	1.4×10^{-4} (33, -3.82)	27 / 6 / 27
B2 ridge regression	1.773	1.486–2.042	1.353	-0.124	1.735	1.872	-0.039	$[-0.428, +0.311]$	0.369 (60, -0.90)	40 / 20 / 0
B3 language model (first run)	1.307	1.047–1.559	0.967	-0.250	1.430	0.884	$+0.427$	$[+0.236, +0.601]$	0.030 (54, $+2.17$)	25 / 29 / 6
B2-GBM (control)	1.725	1.409–2.032	1.274	$+0.033$	1.595	2.041	$+0.009$	$[-0.409, +0.362]$	0.591 (60, -0.54)	37 / 23 / 0
ZHIZHI, preregistered	1.734	1.372–2.065	1.152	-0.714	1.802	1.532	—	—	—	—

(b) Leave-one-out on the earlier sets

Method	All 699 training cards	Same 481 cards as ZHIZHI	ΔRMSE, 481 cards [90%]	p_{W}
B0 constant	2.893	2.629	-1.088 $[-1.252, -0.918]$	1.5×10^{-32}
B1 type × might	2.389	2.032	-0.491 $[-0.613, -0.366]$	5.5×10^{-23}
B2 ridge regression	1.831	1.568	-0.027 $[-0.162, +0.106]$	0.004
B3 language model (first run)	—	—	—	—
B2-GBM (control)	1.852	1.529	$+0.012$ $[-0.138, +0.162]$	0.106
ZHIZHI, preregistered	—	1.541	—	—

(a) Mean res.: mean residual, predicted − printed cost; units / spells: RMSE on the 44 units and the 16 spells; "$+/-/=$": cards on which ZHIZHI's squared error is smaller than, larger than, or equal to the baseline's. p_{W}: Wilcoxon signed-rank on paired squared errors, zero differences dropped (n remaining), tie-corrected normal approximation (z). B2-GBM is a gradient-boosting control on the same 49 features (depth 2, 256 trees, learning rate 0.05, at least 5 cards per leaf; depth and tree count chosen by ten-fold cross-validation over depths 2–4 and 1–1000 trees); it is reported for comparison only. (b) Leave-one-out refits each baseline per held-out card on the 699 units and spells of the four earlier sets and *Proving Grounds*; the 481-card column uses the cards predictable under the ZHIZHI leave-one-out of Section VII-B, and ΔRMSE compares the nominal ZHIZHI leave-one-out (1.541) with the baseline on those cards. B3 was not run on the earlier sets. Error distribution on the 60 cards (cards with absolute error exactly 0 / >3 C): ZHIZHI 19 / 4, B3 8 / 3.

TABLE S18
THE 49 CARD-FACE FEATURES OF THE RIDGE-REGRESSION BASELINE B2, ALL MACHINE-READ FROM THE CARD FACE; NO HAND-PRICED QUANTITY ENTERS

Group	Columns	Definition
Type	1	Spell $= 1$, unit $= 0$ (the unit gear VEN-058 counts as a unit)
Domain	6	Fury, Calm, Mind, Body, Order, Chaos, one-hot; a two-domain card sets both columns
Rarity	5	Common, uncommon, rare, epic, alternate art, one-hot
Might	1	The unit's might; 0 for spells
Keywords	26	One indicator per keyword occurring on at least three training cards (training-set card counts in parentheses): ACTION (85), REACTION (84), DEFLECT (40), HIDDEN (37), ASSAULT (33), GANKING (33), EMPOWER (30), ACCELERATE (26), TANK (23), DEATHKNELL (23), SHIELD (22), EMPOWERED (22), REPEAT (20), FLOW (17), TEMPORARY (15), AMBUSH (14), HUNT (12), LEVEL (12), WEAPONMASTER (11), ADD (9), VISION (8), LEGION (8), MIGHTY (7), BUFF (6), BURN (5), PREDICT (3)
Rules-text length	1	Characters after removing reminder text (keyword explanations in parentheses) and whitespace; for the six cards available only in English, multiplied by the Chinese-to-English character ratio 0.6054
Sentence count	1	Segments after removing reminder text, split at sentence punctuation, line breaks and bullets, counting only segments with content beyond keyword tokens and punctuation
"Whenever" and "if" counts	2	Literal occurrences (whenever, if on the English cards)
Number count and sum	2	Arabic numerals after removing reminder text (including the 2 of "Assault 2" and cost symbols); Chinese numerals not counted
"Enemy / friendly / all / each" counts	4	Literal occurrences (enemy or enemies, friendly, all, each on the English cards)

Features are standardised on the 699 training cards; the ridge penalty $\lambda = 1.259$ was chosen by ten-fold cross-validation (seed 20260924) over 61 values from 10^{-2} to 10^{4}, with ten-fold RMSE 1.833 C. The largest standardised coefficients are might $+2.766$, spell $+2.072$, epic $+0.568$, REACTION -0.526, Body -0.507 and common -0.505. DEPLOY, which occurs on two evaluation cards and on no training card, is not a feature.

TABLE S19
COUNT-DEPENDENT EFFECTS: IMPLIED NUMBER OF INSTANCES N^* (CONVENTION 1). (A) BONUS OR COST REDUCTION PER X, 24 CARDS; (B) DAMAGE TO EVERY ENEMY UNIT, 7 CARDS

(a) Bonus or cost reduction per X

Card	Class	Counted object	C	F	Per instance	N^*
High Note (RAD-048)	bonus per X	exhausted units	3	0	1	3.00
Seraphine, Not Alone (RAD-138)	bonus per X	other exhausted units	7	2.4	1	4.60
Party Crasher (RAD-122)	reduction per X	exhausted units	11	6	1	5.00
Friendship (UNL-046)	bonus per X	distinct tags	1	0	1	1.00
Daisy! (UNL-196)	reduction per X	distinct tags	13	10	1	3.00
Scrutinizing Sergeant (UNL-157)	bonus per X	other friendly units	6	6.4	0.4	-1.00
Petal Pixie (UNL-076)	bonus per X	TEMPORARY allies here	2	2	1	0.00
Illaoi, Prophet of the Great Kraken (VEN-109)	bonus per X	token units	6	5.4	1	0.60
Sett, Kingpin (OGN-240)	bonus per X	buffed allies here (itself included)	6	5	1	1.00
Kadregrin the Infernal (OGN-038)	bonus per X	other MIGHTY units	13	11	2	1.00
Spiderling (VEN-097)	bonus per X	other units here with the same name	3	1	1	2.00
Show of Strength (SFD-106)	bonus per X	MIGHTY units	4	0	2	2.00
Jaull-Fish (SFD-103)	reduction per X	MIGHTY units	11	6	2	2.50
Shadowcloaked Lurker (VEN-096)	reduction per X	copies of itself in the trash	5	5	2	0.00
Rhasa the Sunderer (OGN-195)	reduction per X	cards in the trash	12	6	1	6.00
Bloodscent Swarm (RAD-080)	reduction per X	battlefields conquered this turn	12	8	2	2.00
Right of Conquest (UNL-015)	bonus per X	battlefields controlled	5	2	2	1.50
Needlessly Large Yordle (SFD-055)	reduction per X	battlefields held this turn	16	7.5	4	2.12
Against the Odds (SFD-001)	bonus per X	enemy units here	2	0	2	1.00
Ancient Warmonger (SFD-131)	bonus per X	enemy units here	5	4	0.5	2.00
Battering Ram (SFD-012)	reduction per X	cards played this turn	5	5	1	0.00
Repair Specialist (VEN-076)	bonus per X	gear	3	3	0.5	0.00
Plaza Guardian (VEN-064)	reduction per X	gear	10	8	1	2.00
Ornn, Forge God (SFD-085)	bonus per X	gear	6	4	1	2.00

(b) Damage to every enemy unit (value per target = damage per unit)

Card	Type	C	F	Per target	N^* (1:1)	N^* ($B(c)$ on units)	Reading
Cannon Barrage (OGN-127)	Spell	4	0	2	2.00	2.00	REACTION; 2 damage to every enemy unit in the showdown (v6.1: deferred, all targets)
Firestorm (OGS-002)	Spell	8	0	3	2.67	2.67	3 damage to every enemy unit at one battlefield (v6.1: deferred)
Bullet Time (OGN-268)	Spell	2	0	1	2.00	2.00	Priced at the margin (first author's reading): each extra rune paid (2 C) adds 1 damage to every enemy unit at one battlefield; the base 1 C is an entry price
Bellows Breath (SFD-080)	Spell	3	0	1	3.00	3.00	ACTION, REPEAT at 0; 1 damage to each of up to three units at one location (v6.1 booked one target at 1.00; here the whole area part is taken out)
Kog'Maw, Caustic (OGN-190)	Unit	5	1	2	2.00	2.25	1 might; DEATHKNELL: 4 damage to every unit at its battlefield, $\times 0.5 = 2$ per enemy unit (own damage not counted)
Anivia, Primal (OGN-148)	Unit	11	8	3	1.00	1.83	8 might; when attacking, 3 damage to every enemy unit here (repeatable; booked once)
Graves, Blasting Through (RAD-085)	Unit	7	0	4	1.75	2.04	4 might $-$ 4 for "can't deal combat damage" (first author's reading) $= 0$; when attacking or defending, damage equal to its might (4) to every enemy unit here, booked once

(a) Median N^* 2.00 over the 24 cards (95% bootstrap interval 1.00–2.06); earlier sets alone ($n = 20$) 1.25 (0.80–2.00), mean 1.44; fifth set: Party Crasher 5.00, Bloodscent Swarm 2.00, Seraphine, Not Alone 4.60, High Note 3.00. By counted object: exhausted units 4.60 ($n = 3$), distinct tags 2.00 (2), own units of a kind 1.00 (8), cards in the trash 3.00 (2), battlefields 2.00 (3), enemy units 1.50 (2), cards played this turn 0.00 (1), gear 2.00 (3). Leave-one-out prediction $\hat{C} = F + N\times$(per instance) over the 24 cards, RMSE in C: $N = 0$ (v6.1 nominal) 3.304; $N = 0.5$ 2.676; $N = 1$ 2.146; $N = 2$ 1.761; $N = 3$ 2.530; $N =$ leave-one-out median 1.761 (the leave-one-out median is 2.00 for every card). Paired ΔRMSE against $N = 0$, 90% interval: $N = 2$ -1.543 (-2.450 to -0.689, Wilcoxon $p = 0.010$); $N = 3$ -0.774 (-1.720 to $+0.208$, $p = 0.281$). Out of sample, with N fixed on the earlier sets only (median 1.25) the four fifth-set cards have RMSE 2.766 (mean residual -2.59), against 4.218 (-4.15) at $N = 0$, 2.047 (-1.65) at $N = 2$ and 1.625 (-0.40) at $N = 3$.

(b) Median N^* 2.00 at 1:1 and 2.04 with the budget curve on units. A separate check on 14 area-damage spells run on 6 October, before the costs were read, gave median 2.0 (95% interval 1.30–2.45). Over all 31 cards of (a) and (b): median N^* 2.00 (quartiles 1.00–2.31; units 2.00, spells 2.00) at 1:1, with leave-one-out RMSE 1.698 (units 1.876, spells 1.155); 2.25 (1.46–2.92; units 2.34, spells 2.00) with the budget curve on all units, leave-one-out RMSE 1.733 (units 1.883, spells 1.296); ΔRMSE $+0.035$ (-0.182 to $+0.211$), Wilcoxon $p = 0.445$. The two symmetric-damage spells Unchecked Power and Avalanche are in Table S21.

TABLE S20
OPTIONAL ADDITIONAL COST BUYING AN ON-PLAY EFFECT (CONVENTION 5): PRINTED COST C, VALUE OF THE UNPAID VERSION F, COST OF THE OPTION K, VALUE OF THE EFFECT IT BUYS E

Card	C	F	K	E	$E-K$	$C-F$
Clockwork Keeper (OGN-044)	2	2	2	2	$+0.00$	$+0$
Blast Corps Cadet (SFD-013)	2	2	3	2	-1.00	$+0$
Frostcoat Cub (SFD-067)	3	3	2	1	-1.00	$+0$
Sea Monkey (SFD-098)	2	2	1	1	$+0.00$	$+0$
Zaun Punk (SFD-160)	3	3	1	2.75	$+1.75$	$+0$
Pyke, Dockside Butcher (UNL-028)	3	3	2	3	$+1.00$	$+0$
Nami, Headstrong (UNL-052)	3	3	2	2	$+0.00$	$+0$
Masa, Crashing Thunder (VEN-120)	4	4	2	2	$+0.00$	$+0$
Zed, From the Shadows (VEN-023)	6	4	1	3.9	$+2.90$	$+2$
Ruthless Strike (VEN-008)	3	3	1	2	$+1.00$	$+0$
Meditation (OGN-048)	2	2	1	2	$+1.00$	$+0$
Crescent Guardian (UNL-122)	4	4	2	2	$+0.00$	$+0$
Shimmering Swimmer (RAD-014)	3	3	1	2	$+1.00$	$+0$

$C-F=0$ on 12 of 13 cards; the exception is Zed, From the Shadows ($+2$). Net value of the paid option: median $+0.00$, range -1.00 to $+2.90$.

TABLE S21
SYMMETRIC KILL OR DAMAGE (CONVENTION 3): ENEMY LOSSES AT FULL PRICE, OWN LOSSES AT ONE HALF, DEFAULT TWO ENEMY UNITS AND ONE OWN

Card	C	Reading	Predicted	Predicted $- C$	v6.1 value
Cull the Weak (OGN-209)	4	Each player chooses and kills one of their own units: enemy 1×8, own $1 \times (-4)$	4	$+0$	deferred
Safety Inspector (UNL-164)	7	3 might; on play each player kills one of their own units (optional XP payment not taken): enemy 1×8, own $1 \times (-4)$	7	$+0$	3.00
King's Edict (OGN-237)	10	One-on-one: the opponent chooses one of their units and kills it: enemy 1×8	8	-2	1.40 (low confidence)
Unchecked Power (OGN-123)	11	At the battlefield, enemy 2×8 by default, own $1 \times (-4)$, plus one own unit exhausted (-1); played only when favourable	11	$+0$	deferred
Unchecked Power (OGN-123)	11	As above, but two units on each side	8	-3	deferred
Tibbers (OGS-018)	12	7 might; on play 3 damage to every unit at battlefields: enemy 2×3, own $1 \times (-1.5)$ (own damage halved)	11.5	-0.5	7.00 (deferred)
Flurry of Blades (OGN-133)	1	1 damage to every unit at battlefields: enemy 2×1, own $1 \times (-0.5)$	1.5	$+0.5$	deferred
Avalanche (RAD-079)	9	3 damage to every unit in every base: enemy 2×3, own $1 \times (-1.5)$ (same reading)	4.5	-4.5	deferred
Avalanche (RAD-079)	9	As above, but three enemy units and no own loss	9	$+0$	deferred

v6.1 booked symmetric effects at zero (a card consisting only of them was deferred) and an opponent-chosen kill of their own unit at 1.40 C (Table S1), giving Safety Inspector 3 against 7 C and King's Edict 1.4 against 10 C. The reading of own losses at -4 C (half of an unconditional kill) and the two-enemy, one-own default for Unchecked Power were fixed by the first author after the fifth-set costs had been seen; the -4 C is pinned exactly by Cull the Weak and Safety Inspector, which kill one unit on each side and need no count default. Alternative readings of Unchecked Power, as the implied number of units killed on each side (the same number on both sides): own kill -2 (v6.1 additional-cost price) and no exhaust 1.83; own kill -2 with one own unit exhausted (-1) 2.00; own kill -4 2.75; own kill -5 (the sacrifice reading of Table S5) 3.67; own losses ignored 1.38. Killing a friendly unit as an additional cost remains at -2 C (Convention 4).

TABLE S22
(A) OPTIONAL COST REDUCTION ASSUMED TAKEN (CONVENTION 6); (B) OPTIONAL ON-PLAY PAYMENT BOOKED AT ONE HALF (CONVENTION 7)

(a) "You may pay x as an additional cost; if you do, my cost is reduced by r"

Card	C	F	r	x	$C - F$	$F + r - x$	$F + r$
Brazen Buccaneer (OGN-002)	6	5	2	1	$+1$	6	7
Poppy, Defender of the Meek (UNL-178)	8	5	3	1.5	$+3$	6.5	8
Skystrike Squadron (RAD-075)	5	4	2	1	$+1$	5	6
Dune Devourer (RAD-125)	9	7	2	2	$+2$	7	9

(b) "When you play me, you may pay x to …"; $f^* = (C - F)/$net value

Card	C	F	Net value	f^*	$\hat{C}$ at $f = 0$ / 0.5 / 1
Wildclaw Shaman (OGN-147)	4	3	2	0.50	3 / 4 / 5
Kennen, Keeper of Balance (VEN-135)	3	2	0	— (net value 0)	2 / 2 / 2
Mordekaiser, Dominating (RAD-137)	13	8	7	0.71	8 / 11.5 / 15

(a) RMSE of the predicted cost over the 4 cards (earlier sets 2 / fifth set 2): F alone 1.936 (2.236 / 1.581); $F + r - x$ 1.250 (1.061 / 1.414); $F + r$ 0.707 (0.707 / 0.707). The two rules cannot be separated on the earlier sets alone (two cards each); both are carried in Table S25, and the main chain uses $F + r$.
(b) RMSE over the 3 cards at $f = 0$ / 0.5 / 1: 3.000 / 1.041 / 1.414 (earlier-set cards alone 1.000 / 0.707 / 1.000; Mordekaiser, Dominating residual -5.00 / -1.50 / $+2.00$). Kennen, Keeper of Balance was moved out of the evidence on independent re-check: the on-play payment itself has net value 0.

TABLE S23
CONDITIONAL COST REDUCTION (CONVENTION 8): FRACTION OF THE REDUCTION r THAT THE PRINTED COST CREDITS, $f^* = (C - F)/r$

Card	Type	C	F	r	f^*
Void Drone (SFD-010)	Unit	3	3	2	$+0.00$
Monch (UNL-035)	Unit	6	6	2	$+0.00$
Noxus Hopeful (OGN-012)	Unit	4	4	2	$+0.00$
Keeper of Law (VEN-119)	Unit	7	5	4	$+0.50$
Drag Under (SFD-164)	Spell	7	8	2	-0.50
Hold 'Em (RAD-070)	Spell	3	3	2	$+0.00$
Production Surge (SFD-076)	Spell	6	5.4	2	$+0.30$
Shock Blast (VEN-059)	Spell	5	4	2	$+0.50$
Spoils of War (OGN-144)	Spell	6	4	2	$+1.00$

Units: median f^* $+0.00$ ($n = 4$; earlier sets alone $+0.00$). Spells: median $+0.30$ ($n = 5$; earlier sets alone $+0.40$). v7 does not credit the reduction on units and leaves spells unchanged (evidence too thin); as a sensitivity, booking spells at the earlier-set median 0.40 moves the fifth-set error from 1.100 to 1.105 C (Table S25) and, at the leave-one-out median on the 4 earlier-set spells, leaves the 481-card leave-one-out error at 1.274 C.

TABLE S24
WHERE THE MIGHT BUDGET $B(c)$ ENTERS (CONVENTION 9): IMPLIED COUNT AND PREDICTION ERROR UNDER FOUR CONVENTIONS. SPELLS ARE ALWAYS 1:1

	Count-dependent and area cards ($n = 31$)			Earlier sets, LOO ($n = 481$)		Fifth set ($n = 60$)	
Convention	N^* median (U / S)	Quartiles	LOO RMSE (U / S)	Class cards	All units	Class cards	All units
Everything 1:1	2.00 (2.00 / 2.00)	1.00–2.31	1.698 (1.876 / 1.155)	1.274	1.274	1.100 ($N = 2.00$)	1.100
$B(c)$ on all units	2.25 (2.34 / 2.00)	1.46–2.92	1.733 (1.883 / 1.296)	1.279	1.415	1.126 ($N = 2.25$)	1.229
$B(c)$ on units $\geq$ 7 C only	2.00 (2.00 / 2.00)	1.00–2.62	1.631 (1.790 / 1.155)	1.274	1.326	1.109 ($N = 2.00$)	1.163
$B(c)$ on units $\geq$ 6 C only	2.00 (2.02 / 2.00)	1.46–2.78	1.710 (1.890 / 1.155)	1.278	1.358	1.138 ($N = 2.00$)	1.211

U / S: units / spells. "Class cards": only the count-dependent and area cards are re-read under the convention; "all units": every unit's predicted cost is read off the curve, $\hat{C} = G^{-1}(V)$. $B(c)$ is used only to separate effect value from might when a convention is inverted on a unit; "$\geq$ 7 C only" keeps 1:1 below 6 C and joins the two linearly between 6 and 7 C (vanilla 6 might at 6 C, 8 at 7 C). The first row is the full v7 rule set (S7 of Table S25; 1.274 on the 481 cards, 1.100 on the fifth set). Paired ΔRMSE against "everything 1:1", 90% intervals, in the order class leave-one-out; 481 cards, class only; 481 cards, all units; fifth set, all units. $B(c)$ on all units: $+0.035$ (-0.182 to $+0.211$); $+0.006$ (-0.005 to $+0.018$); $+0.142$ ($+0.088$ to $+0.199$); $+0.129$ (-0.029 to $+0.282$). Units $\geq$ 7 C only: -0.067 (-0.278 to $+0.109$); $+0.000$ (-0.008 to $+0.011$); $+0.052$ (-0.006 to $+0.112$); $+0.062$ (-0.081 to $+0.214$). Units $\geq$ 6 C only: $+0.011$ (-0.217 to $+0.192$); $+0.004$ (-0.006 to $+0.017$); $+0.084$ ($+0.027$ to $+0.143$); $+0.111$ (-0.046 to $+0.263$).

TABLE S25
FIFTH SET, 60 PRIMARY CARDS: ERROR OF ZHIZHI AS THE CANDIDATE CONVENTIONS ARE ADDED ONE AT A TIME, AND PAIRED DIFFERENCE FROM THE LANGUAGE-MODEL BASELINE B3 (ΔRMSE = ZHIZHI $-$ B3; NEGATIVE FAVOURS ZHIZHI)

			vs. B3, first run (1.307)		vs. B3, five-run mean (1.217)	
Version	RMSE	Mean residual	ΔRMSE [90%]	p_{W}	ΔRMSE [90%]	p_{W}
S0 ZHIZHI, nominal (preregistered)	1.734	-0.71	$+0.427$ [$+0.236$, $+0.601$]	0.030	$+0.516$ [$+0.316$, $+0.702$]	0.014
S1 v6.1 AMBER convention (repeatable triggers at one half, a quarter under an independent condition)	1.513	-0.35	$+0.206$ [$+0.015$, $+0.384$]	0.179	$+0.295$ [$+0.089$, $+0.479$]	0.140
S2 + Section IX revisions (Hunker Down, RAD-034: "can't move" at -2; Fanfare, RAD-047: capped repeat count at $k = 0.5$)	1.445	-0.32	$+0.138$ [-0.047, $+0.295$]	0.351	$+0.228$ [$+0.027$, $+0.405$]	0.351
S3 + count default $N = 1.25$ (fixed on the earlier sets only)	1.209	-0.21	-0.098 [-0.286, $+0.090$]	0.843	-0.008 [-0.193, $+0.165$]	0.707
S4 + optional cost reduction, rule $F + r - x$	1.202	-0.19	-0.105 [-0.294, $+0.081$]	0.744	-0.015 [-0.198, $+0.160$]	0.831
S4′ + optional cost reduction, rule $F + r$ (main chain)	1.181	-0.14	-0.126 [-0.308, $+0.063$]	0.572	-0.036 [-0.218, $+0.138$]	0.898
S6 = S4′ + count default $N = 2$ + "up to N" at $\min(N, 2)$ (reverse-engineered)	1.095	-0.08	-0.212 [-0.434, $+0.014$]	0.417	-0.122 [-0.333, $+0.076$]	0.659
S7 = S6 + stun price staged by set (Section IX) = all v7 rules	1.100	-0.10	-0.207 [-0.431, $+0.018$]	0.419	-0.117 [-0.334, $+0.083$]	0.614
Sensitivity: S7 + spell conditional cost reduction at $f = 0.40$ (earlier-set spell median; not adopted)	1.105	-0.09	-0.202 [-0.425, $+0.024$]	0.505	-0.112 [-0.329, $+0.090$]	0.712
Sensitivity: S7 with $N = 3$	1.071	-0.02	-0.236 [-0.474, $+0.005$]	0.347	-0.146 [-0.370, $+0.070$]	0.498
Sensitivity: S7 with Mordekaiser, Dominating booked as firing (on-play payment once, in full: 15)	1.114	-0.04	-0.193 [-0.415, $+0.027$]	0.419	-0.104 [-0.314, $+0.092$]	0.619
Pure out-of-sample: S1 + $N = 1.25$ + rule $F + r$ (every parameter from the earlier sets; without the Section IX revisions)	1.263	-0.18	-0.044 [-0.258, $+0.167$]	0.928	$+0.046$ [-0.157, $+0.239$]	0.493
Pure out-of-sample with $N = 2$ (6 October draft value, before unblinding)	1.181	-0.12	-0.126 [-0.381, $+0.123$]	0.710	-0.036 [-0.273, $+0.188$]	0.908
S7 without the signature spell Last Caress (RAD-154), $n = 59$ (B3 on the same 59 cards: first run 1.316, five-run mean 1.219)	1.079	-0.07	-0.238 [-0.460, -0.007]	0.293	-0.141 [-0.347, $+0.070$]	0.468

RMSE and mean residual of $V - C_{\mathrm{tot}}$ in C. Eleven cards change value between S1 and S7: S2 Hunker Down (RAD-034), Fanfare (RAD-047); S3 Party Crasher (RAD-122), Bloodscent Swarm (RAD-080), Seraphine, Not Alone (RAD-138), High Note (RAD-048); S4 Skystrike Squadron (RAD-075), Dune Devourer (RAD-125); S6 Bilgewater Tracker (RAD-003); S7 Riven, Broken Blade (RAD-043) and RAD-104; the other 49 cards are unchanged. The count default of S3 is fixed on the earlier sets alone; the two optional-cost-reduction rules cannot be separated on the earlier sets (two cards each), so both are listed, and the main chain uses $F + r$; the spell conditional cost reduction is not adopted (v7) and appears only as a sensitivity. Paired bootstrap over cards, 2000 resamples; p_{W} Wilcoxon signed-rank on paired squared errors. The B3 runs are described in Table VI.

TABLE S26
EARLIER SETS, LEAVE-ONE-OUT: THE 37 CARDS WHOSE PREDICTION CHANGES UNDER THE V7 RULES (THE CONVENTIONS AND THE STUN PRICE STAGED BY SET), BY RULE AND CARD BY CARD. PARAMETERS ($\hat{N}$ AND THE ON-PLAY FRACTION) ARE THE LEAVE-ONE-OUT MEDIANS

(a) By rule

Rule	Cards changed	RMSE, v6.1 AMBER (LOO)	RMSE, v7 rules (LOO)	Priced exactly
Count-dependent and area effects (Conv. 1)	19	3.044	1.757	3
"Up to N" targets (Conv. 2)	8	2.272	0.948	3
Symmetric kill (Conv. 3)	1	4.000	0.000	1
Optional cost reduction (Conv. 6)	2	2.236	0.707	1
Optional on-play payment (Conv. 7)	1	1.000	0.429	0
Stun price staged by set (Section IX)	6	0.951	1.819	0

(b) Card by card

Card	C	$\hat{C}$, v6.1 AMBER (LOO)	$\hat{C}$, v7 rules (LOO)	Rule applied
Brazen Buccaneer (OGN-002)	6	5.00	7.00	optional cost reduction, $+2$
Kadregrin the Infernal (OGN-038)	13	12.00	15.00	count-dependent, $\hat{N} = 2.00$
Kinkou Monk (OGN-141)	6	5.00	6.00	up to N targets, $+1$
Wildclaw Shaman (OGN-147)	4	3.00	4.43	on-play optional payment, $\times 0.71$
Udyr, Wildman (OGN-157)	8	7.58	7.22	stun staged by set, -0.36
Rhasa the Sunderer (OGN-195)	12	6.00	8.00	count-dependent, $\hat{N} = 2.00$
Leona, Determined (OGN-238)	6	6.50	5.50	stun staged by set, -1
Sett, Kingpin (OGN-240)	6	5.50	7.00	count-dependent, $\hat{N} = 2.00$
Battering Ram (SFD-012)	5	5.00	7.00	count-dependent, $\hat{N} = 2.00$
Needlessly Large Yordle (SFD-055)	16	7.50	15.50	count-dependent, $\hat{N} = 2.00$
Fae Dragon (SFD-101)	9	8.44	9.44	up to N targets, $+1$
Jaull-Fish (SFD-103)	11	6.00	10.00	count-dependent, $\hat{N} = 2.00$
Ancient Warmonger (SFD-131)	5	4.25	5.00	count-dependent, $\hat{N} = 2.00$
Petal Pixie (UNL-076)	2	2.50	4.00	count-dependent, $\hat{N} = 2.00$
Scrutinizing Sergeant (UNL-157)	6	6.61	7.21	count-dependent, $\hat{N} = 2.00$
Safety Inspector (UNL-164)	7	3.00	7.00	symmetric kill, enemy 8 / own -4
Vi, Peacekeeper (UNL-176)	7	7.00	6.00	stun staged by set, -1
Poppy, Defender of the Meek (UNL-178)	8	5.00	8.00	optional cost reduction, $+3$
Daisy! (UNL-196)	13	11.00	12.00	count-dependent, $\hat{N} = 2.00$
Repair Specialist (VEN-076)	3	3.25	4.00	count-dependent, $\hat{N} = 2.00$
Shadowcloaked Lurker (VEN-096)	5	5.00	9.00	count-dependent, $\hat{N} = 2.00$
Spiderling (VEN-097)	3	1.50	3.00	count-dependent, $\hat{N} = 2.00$
Illaoi, Prophet of the Great Kraken (VEN-109)	6	5.90	7.40	count-dependent, $\hat{N} = 2.00$
Kennen, Keeper of Balance (VEN-135)	3	3.00	2.00	stun staged by set, -1
Singularity (OGN-105)	10	6.00	12.00	up to N targets, $+6$
Back to Back (OGN-206)	3	2.00	4.00	up to N targets, $+2$
Zenith Blade (OGN-262)	7	5.00	3.00	stun staged by set, -2
Flash (OGS-011)	2	1.00	2.00	up to N targets, $+1$
Against the Odds (SFD-001)	2	0.00	4.00	count-dependent, $\hat{N} = 2.00$
Piercing Light (SFD-023)	4	2.00	4.00	up to N targets, $+2$
Bellows Breath (SFD-080)	3	1.00	2.00	count-dependent, $\hat{N} = 2.00$
Show of Strength (SFD-106)	4	0.00	4.00	count-dependent, $\hat{N} = 2.00$
Right of Conquest (UNL-015)	5	2.00	6.00	count-dependent, $\hat{N} = 2.00$
Back Off (UNL-042)	3	4.00	2.00	stun staged by set, -2
Friendship (UNL-046)	1	0.00	2.00	count-dependent, $\hat{N} = 2.00$
Shadows of the Past (VEN-103)	5	2.00	4.00	up to N targets, $+2$
Acceleration Gate (VEN-150)	5	2.00	4.00	up to N targets, $+2$

Leave-one-out RMSE over the 481 predictable cards of Section VII-B: nominal (preregistered) 1.541; every repeatable trigger at one half 1.438; v6.1 AMBER (a quarter under an independent condition) 1.400; the conventions without the staged stun price 1.262; all v7 rules (staged stun price included) 1.274. Paired ΔRMSE of the v7 rules against 1.400 over all 481 cards -0.127 (90% interval -0.239 to -0.033); over the 37 changed cards 2.598 $\rightarrow$ 1.533, -1.065 (-1.824 to -0.322), 19 closer to cost and 12 further, Wilcoxon $p = 0.065$. Every convention lowers the error on the cards it changes; the staged stun price raises it on its six cards. Four cheap units whose might already fills their cost (Petal Pixie, Repair Specialist, Battering Ram, Shadowcloaked Lurker) move away from their cost under the count default; v7 lists them as exceptions, not modelled. Sensitivity: adding the spell conditional cost reduction at its leave-one-out median (4 cards; not adopted) leaves the 481-card error at 1.274.